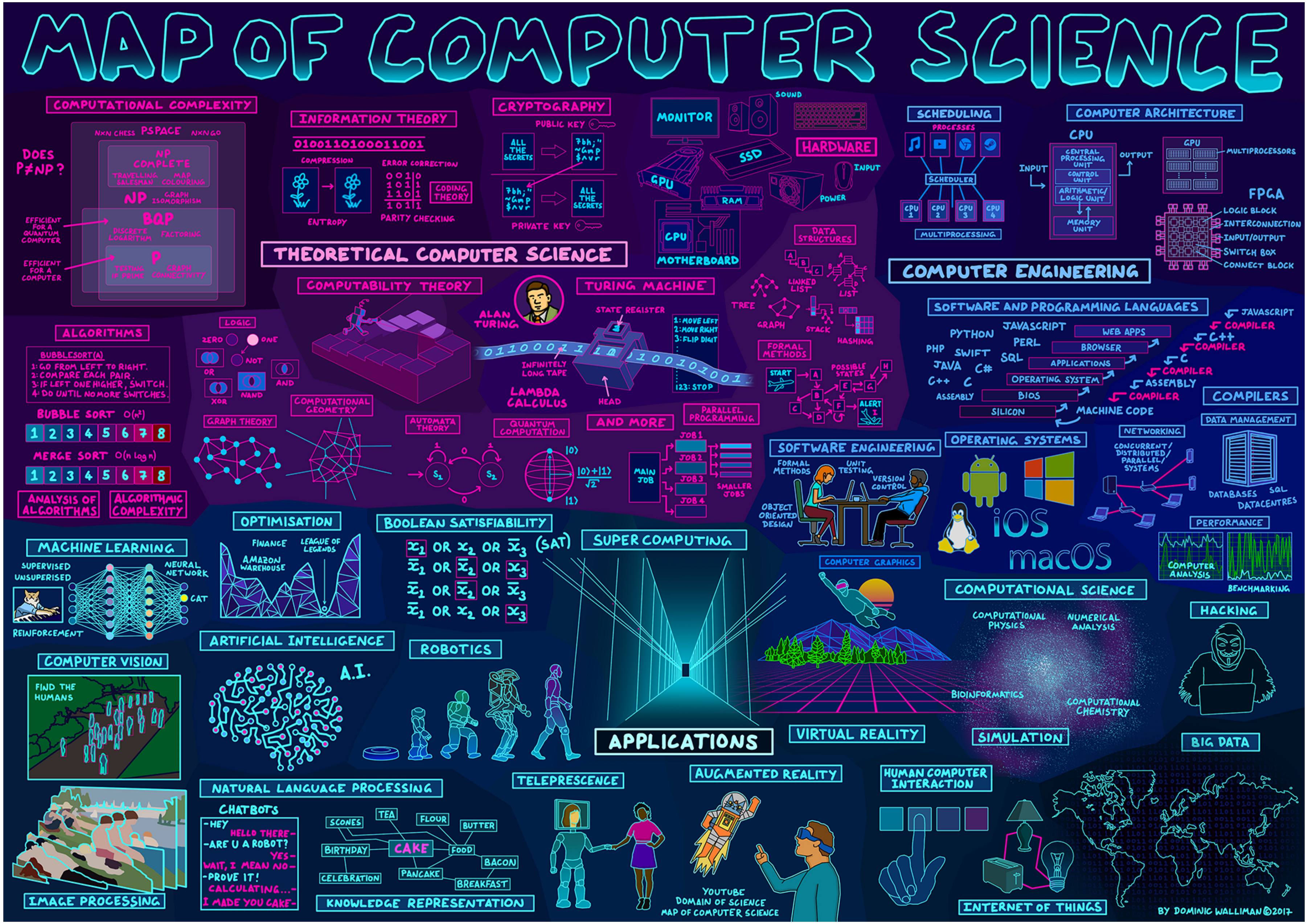
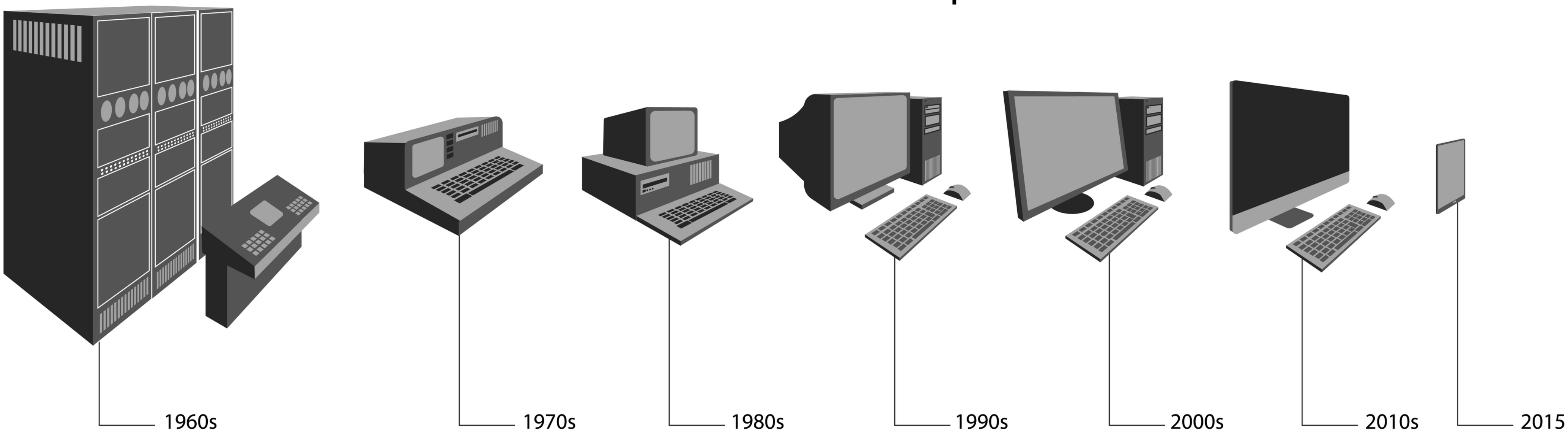
Artificial Intelligence

Dé Céadaoin, 2ú Márta

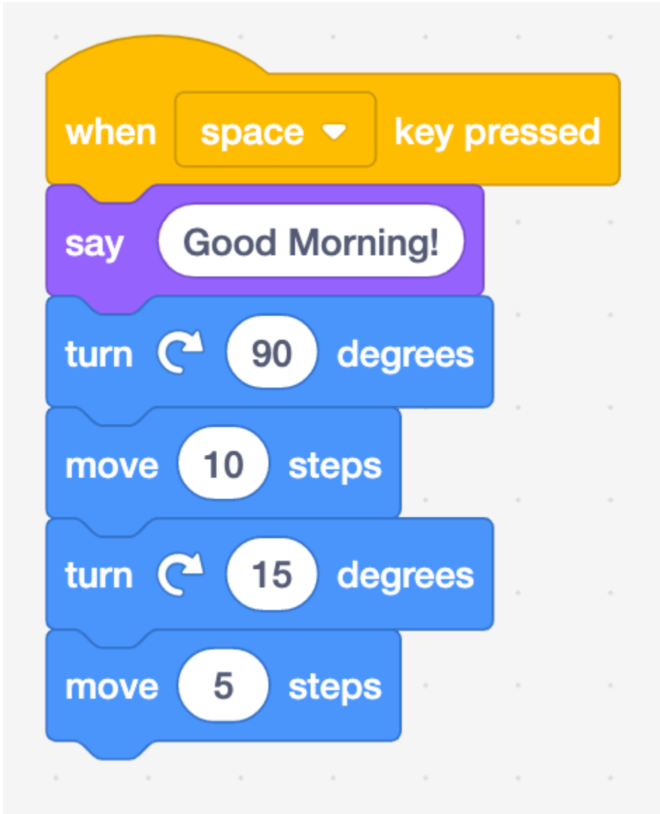

MAP OF COMPUTER SCIENCE
COMPUTATIONAL COMPLEXITY
DOES P≠NP?
PSPACE
NP COMPLETE
TRAVELLING SALESMAN
MAP COLOURING
NP
GRAPH ISOMORPHISM
BQP
DISCRETE LOGARITHM
FACTORING
P
TESTING IF PRIME
GRAPH CONNECTIVITY
EFFICIENT FOR A QUANTUM COMPUTER
EFFICIENT FOR A COMPUTER
INFORMATION THEORY
010011010001100 1
COMPRESSION
ERROR CORRECTION
CODING THEORY
ENTROPY
PARITY CHECKING
CRYPTOGRAPHY
PUBLIC KEY
ALL THE SECRETS
PRIVATE KEY
MONITOR
SOUND
HARDWARE
INPUT
SSD
GPU
RAM
POWER
CPU
MOTHERBOARD
SCHEDULING
PROCESSES
SCHEDULER
MULTIPROCESSING
COMPUTER ARCHITECTURE
CPU
INPUT
OUTPUT
CENTRAL PROCESSING UNIT
CONTROL UNIT
ARITHMETIC/LOGIC UNIT
MEMORY UNIT
GPU
MULTIPROCESSORS
FPGA
LOGIC BLOCK
INTERCONNECTION
INPUT/OUTPUT
SWITCH BOX
CONNECT BLOCK
THEORETICAL COMPUTER SCIENCE
COMPUTER ENGINEERING
COMPUTABILITY THEORY
ALAN TURING
TURING MACHINE
STATE REGISTER
1: MOVE LEFT
2: MOVE RIGHT
3: FLIP DIGIT
123: STOP
INFINITELY LONG TAPE
HEAD
LAMBDA CALCULUS
AND MORE
DATA STRUCTURES
LINKED LIST
LIST
TREE
GRAPH
STACK
HASHING
FORMAL METHODS
START
POSSIBLE STATES
ALERT
PARALLEL PROGRAMMING
MAIN JOB
JOB 1
JOB 2
JOB 3
JOB 4
SMALLER JOBS
SOFTWARE AND PROGRAMMING LANGUAGES
PYTHON
JAVASCRIPT
PHP
SWIFT
PERL
JAVA
C#
SQL
C++
C
ASSEMBLY
WEB APPS
BROWSER
APPLICATIONS
OPERATING SYSTEM
BIOS
SILICON
MACHINE CODE
JAVASCRIPT
COMPILER
C++
COMPILER
C
COMPILER
ASSEMBLY
COMPILER
COMPILERS
ALGORITHMS
BUBBLESORT(A)
1: GO FROM LEFT TO RIGHT.
2: COMPARE EACH PAIR.
3: IF LEFT ONE HIGHER, SWITCH.
4: DO UNTIL NO MORE SWITCHES.
BUBBLE SORT O(n²)
MERGE SORT O(n log n)
ANALYSIS OF ALGORITHMS
ALGORITHMIC COMPLEXITY
LOGIC
ZERO
ONE
NOT
OR
AND
XOR
NAND
COMPUTATIONAL GEOMETRY
GRAPH THEORY
AUTOMATA THEORY
QUANTUM COMPUTATION
|0⟩
|1⟩
(|0⟩+|1⟩)/√2
SOFTWARE ENGINEERING
FORMAL METHODS
UNIT TESTING
VERSION CONTROL
OBJECT ORIENTED DESIGN
OPERATING SYSTEMS
iOS
macOS
NETWORKING
CONCURRENT/DISTRIBUTED/PARALLEL SYSTEMS
DATA MANAGEMENT
DATABASES
SQL
DATACENTRES
PERFORMANCE
COMPUTER ANALYSIS
BENCHMARKING
MACHINE LEARNING
SUPERVISED
UNSUPERVISED
NEURAL NETWORK
CAT
REINFORCEMENT
OPTIMISATION
FINANCE
LEAGUE OF LEGENDS
AMAZON WAREHOUSE
BOOLEAN SATISFIABILITY
x̄₁ OR x₂ OR x̄₃ (SAT)
x̄₁ OR x̄₂ OR x̄₃
x̄₁ OR x̄₂ OR x̄₃
x̄₁ OR x₂ OR x̄₃
SUPER COMPUTING
COMPUTER GRAPHICS
COMPUTATIONAL SCIENCE
COMPUTATIONAL PHYSICS
NUMERICAL ANALYSIS
BIOINFORMATICS
COMPUTATIONAL CHEMISTRY
HACKING
ARTIFICIAL INTELLIGENCE
ROBOTICS
A.I.
COMPUTER VISION
FIND THE HUMANS
APPLICATIONS
VIRTUAL REALITY
SIMULATION
BIG DATA
IMAGE PROCESSING
NATURAL LANGUAGE PROCESSING
CHATBOTS
-HEY
HELLO THERE-
-ARE U A ROBOT?
YES-
WAIT, I MEAN NO-
-PROVE IT!
CALCULATING...-
I MADE YOU CAKE-
SCONES
TEA
FLOUR
BUTTER
BIRTHDAY
CAKE
FOOD
BACON
CELEBRATION
PANCAKE
BREAKFAST
KNOWLEDGE REPRESENTATION
TELEPRESENCE
AUGMENTED REALITY
YOUTUBE
DOMAIN OF SCIENCE
MAP OF COMPUTER SCIENCE
HUMAN COMPUTER INTERACTION
INTERNET OF THINGS
BY DOMINIC WALLIMAN ©2017

Evolution of computers

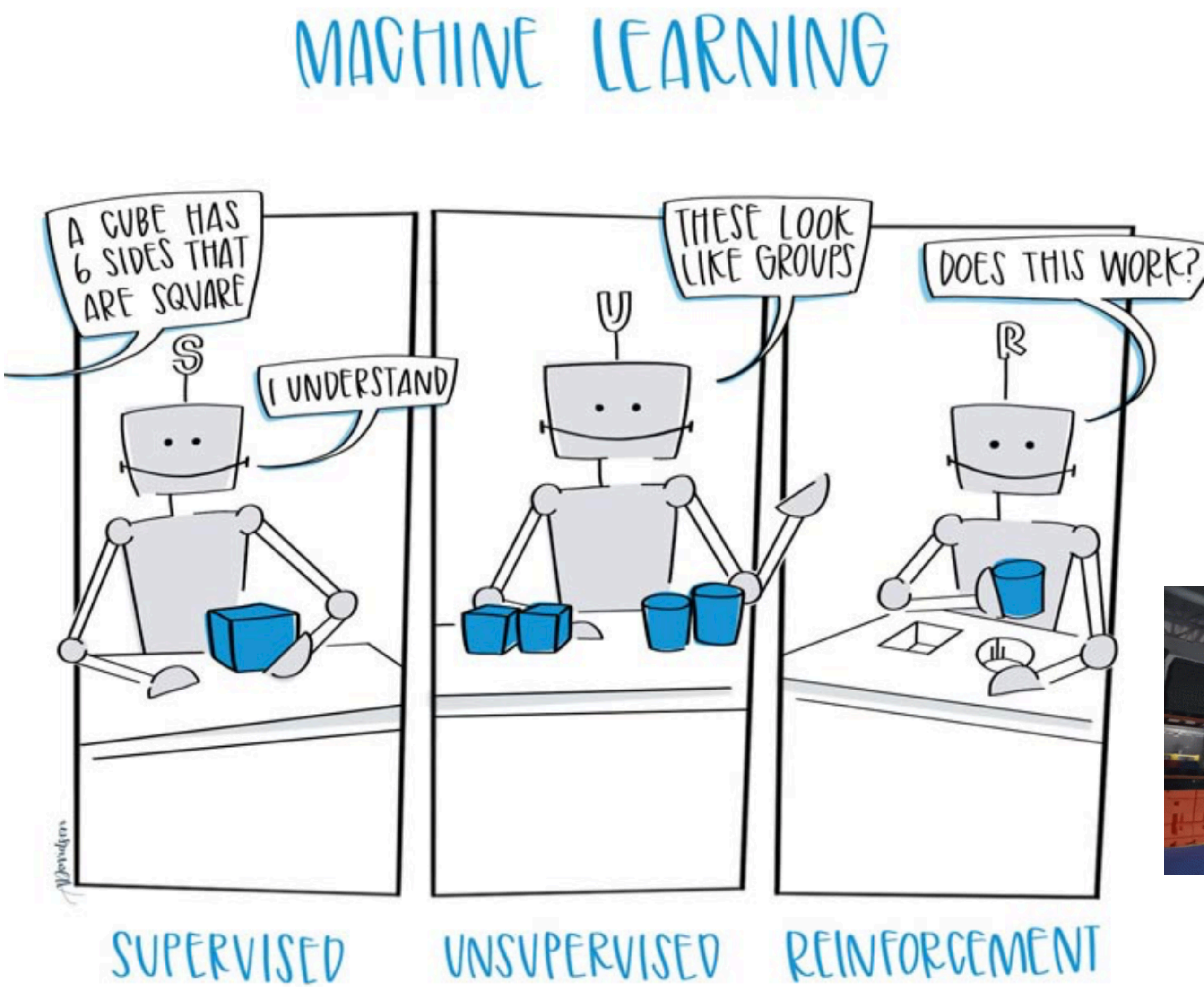


Intelligence?

Morning Routine

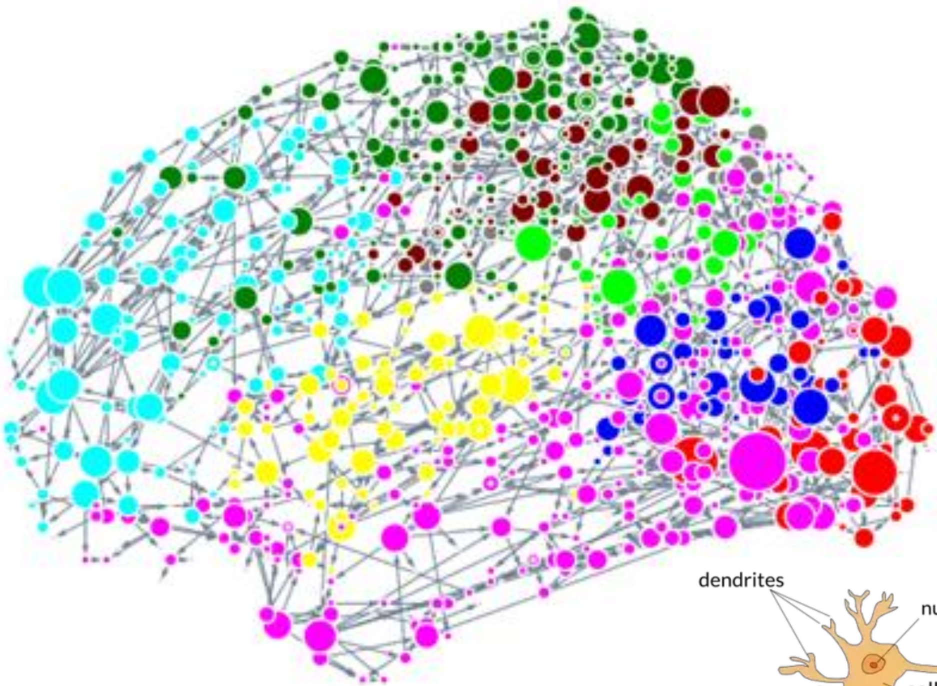


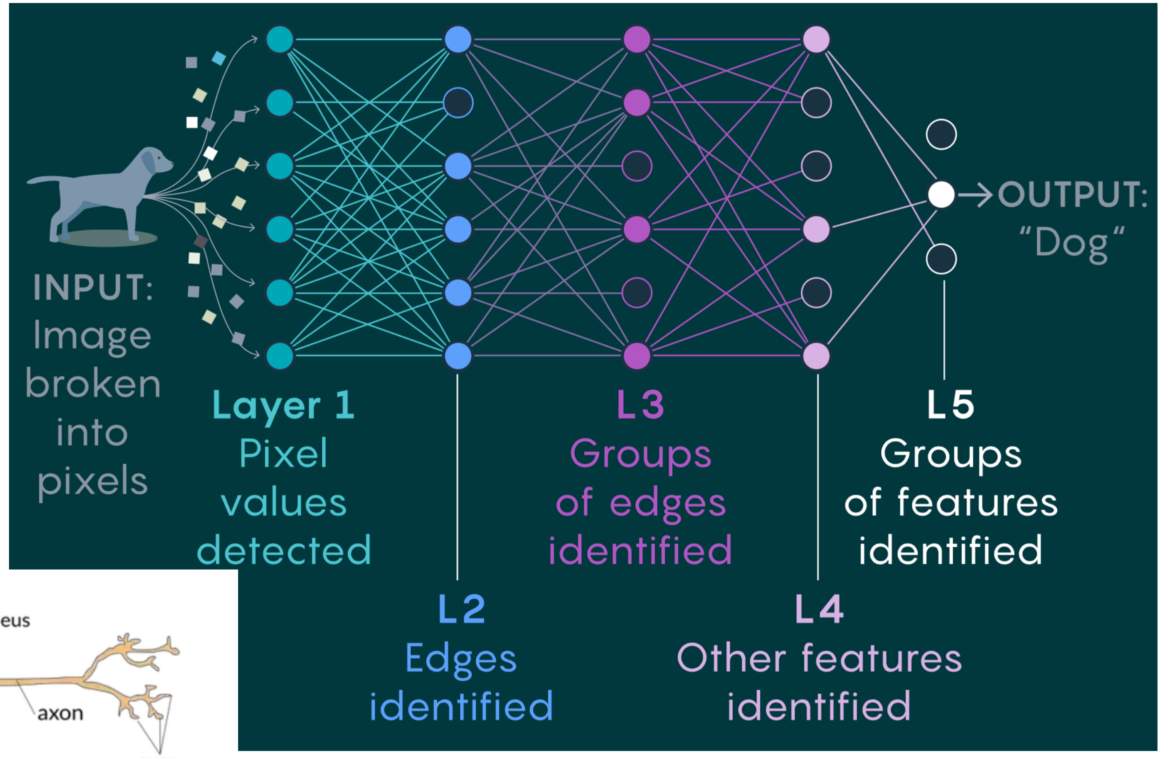
- Walk down stairs
- Find door
- Enter kitchen
- Locate table
- Sit down
- Charge battery

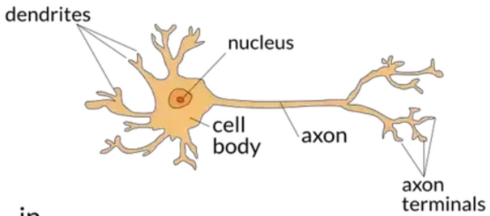

MACHINE LEARNING
A CUBE HAS 6 SIDES THAT ARE SQUARE
I UNDERSTAND
THESE LOOK LIKE GROUPS
DOES THIS WORK?
SUPERVISED
UNSUPERVISED
REINFORCEMENT

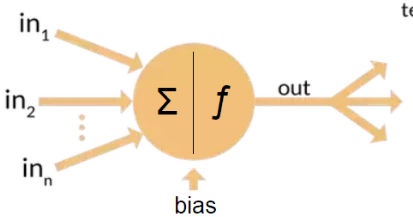

Boston Dynam

Neural Networks

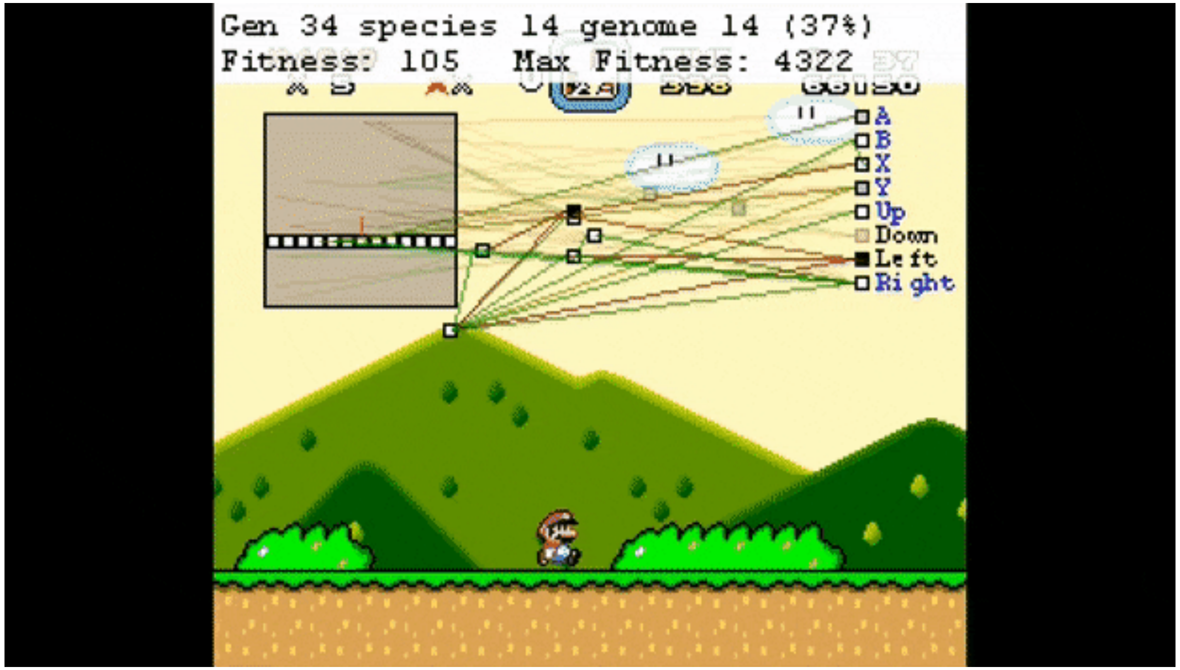


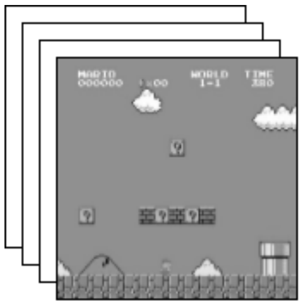


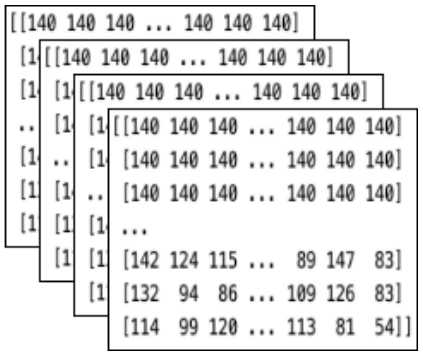


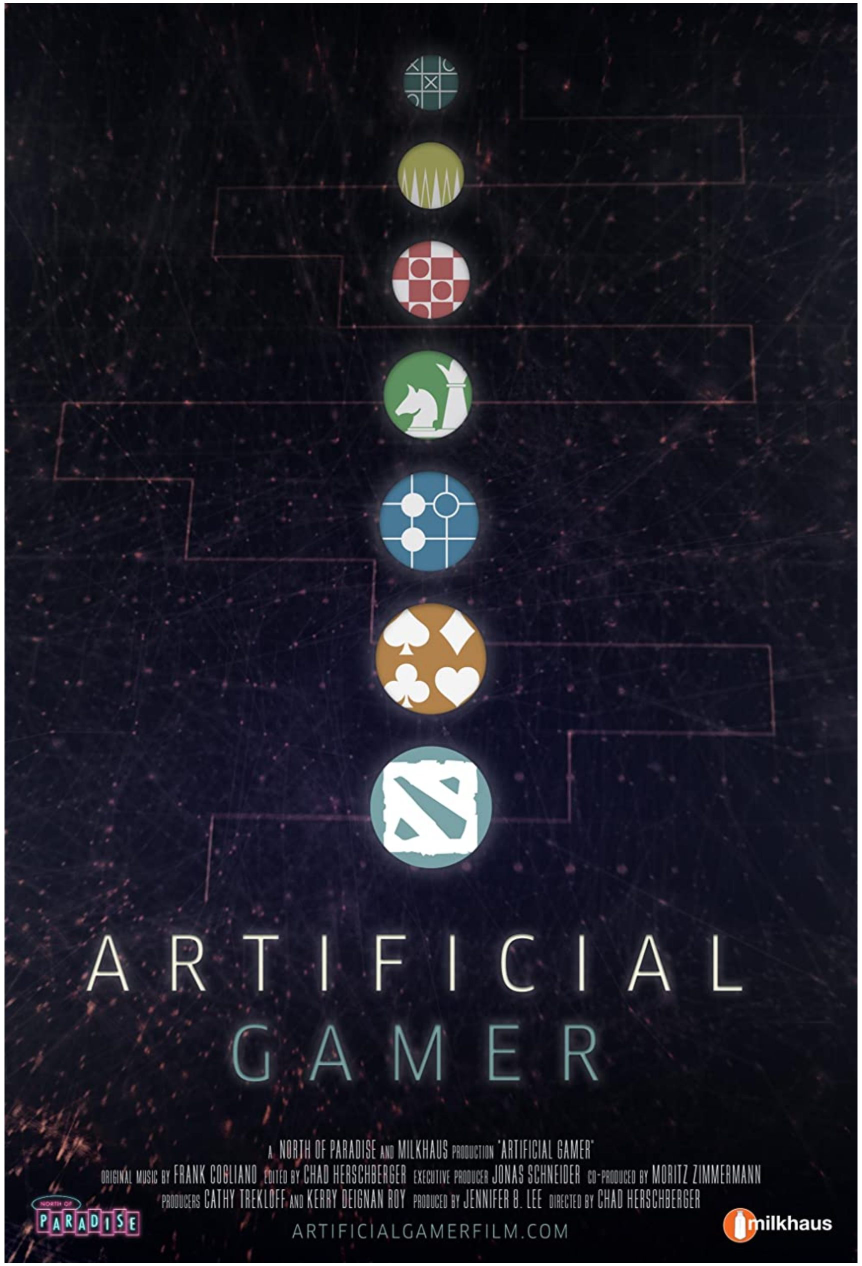


Deep Learning: Try and Try Again





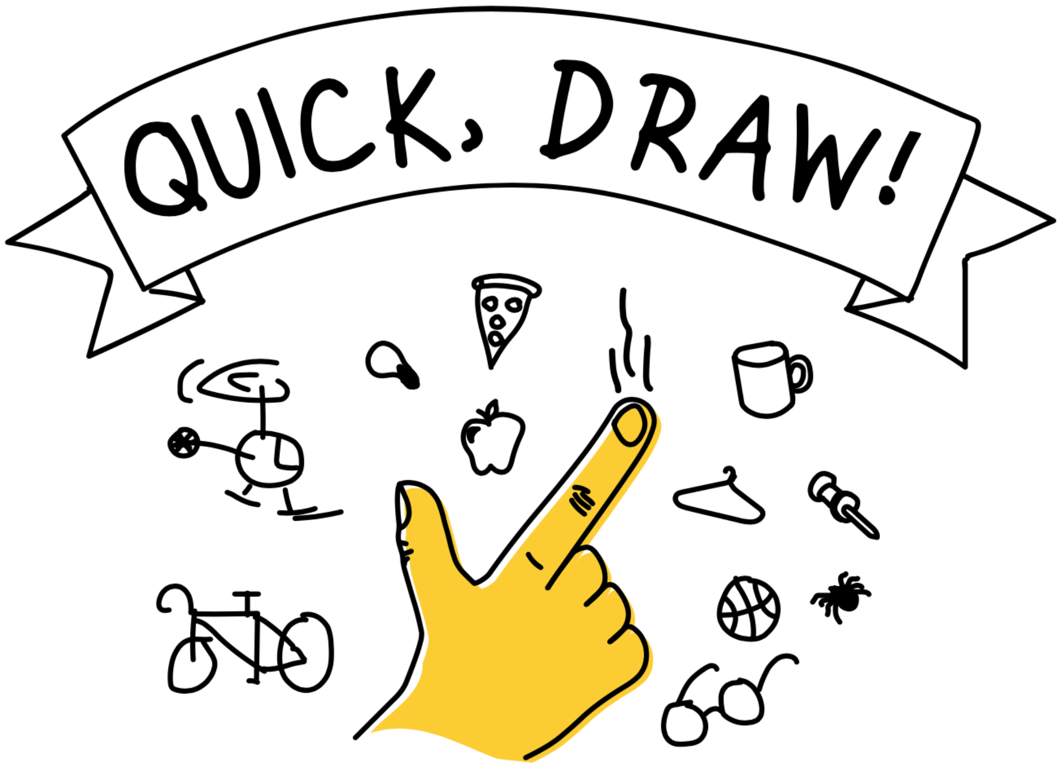



ARTIFICIAL
GAMER
A NORTH OF PARADISE AND MILKHAUS PRODUCTION "ARTIFICIAL GAMER"
ORIGINAL MUSIC BY FRANK COGLIANO EDITED BY CHAD HERSCHBERGER EXECUTIVE PRODUCER JONAS SCHNEIDER CO-PRODUCED BY MORITZ ZIMMERMANN
PRODUCERS CATHY TREKLOFF AND KERRY DEIGNAN ROY PRODUCED BY JENNIFER 8. LEE DIRECTED BY CHAD HERSCHBERGER
NORTH OF PARADISE
ARTIFICIALGAMERFILM.COM
milkhaus


4
9:11
2
Jakiro
Kunkka
Juggernaut
TINY
890 / 1220
116 / 405

SAP Conversational AI

SAP AI Business Services



Can a neural network learn to recognize doodling?

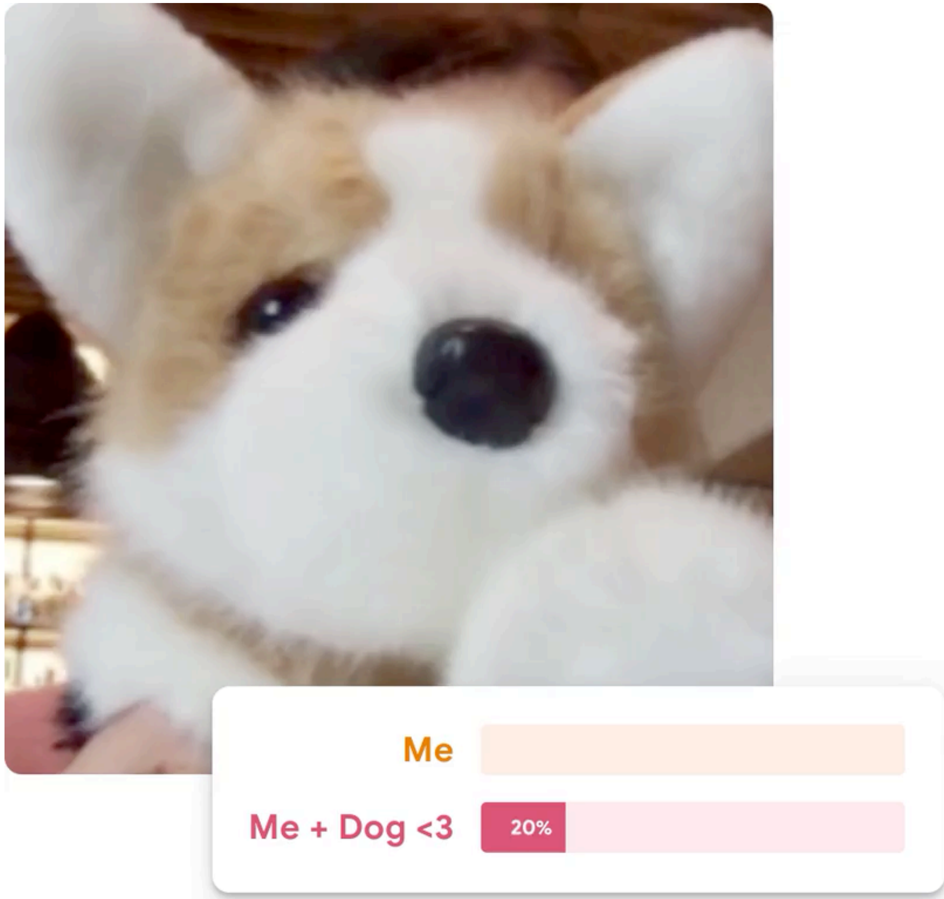
Help teach it by adding your drawings to the world's largest doodling data set, shared publicly to help with machine learning research.

Teachable Machine

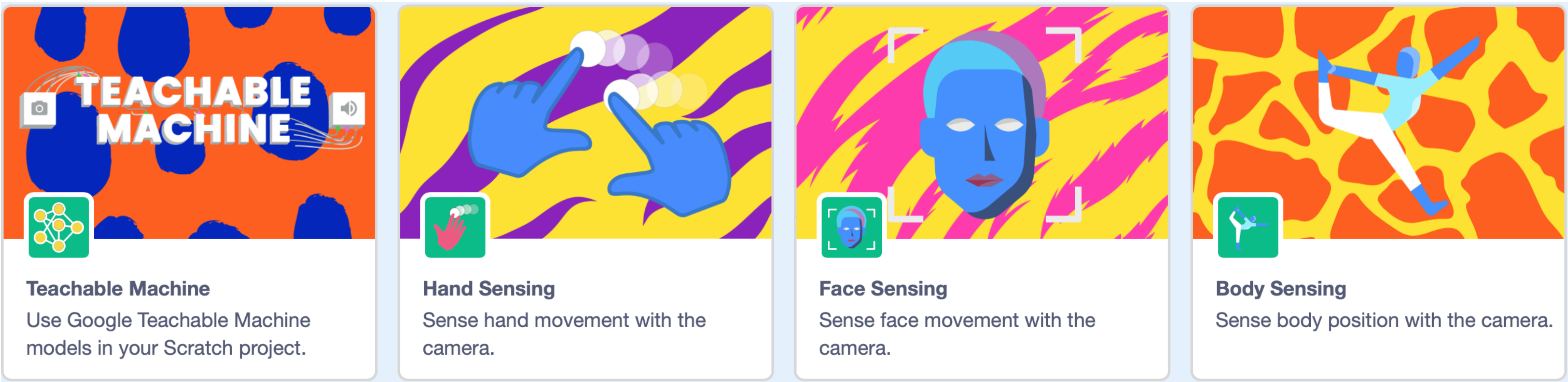
Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



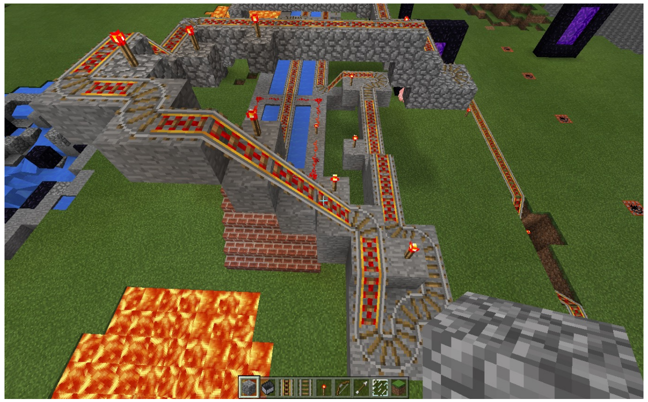
Scratch AI Blocks

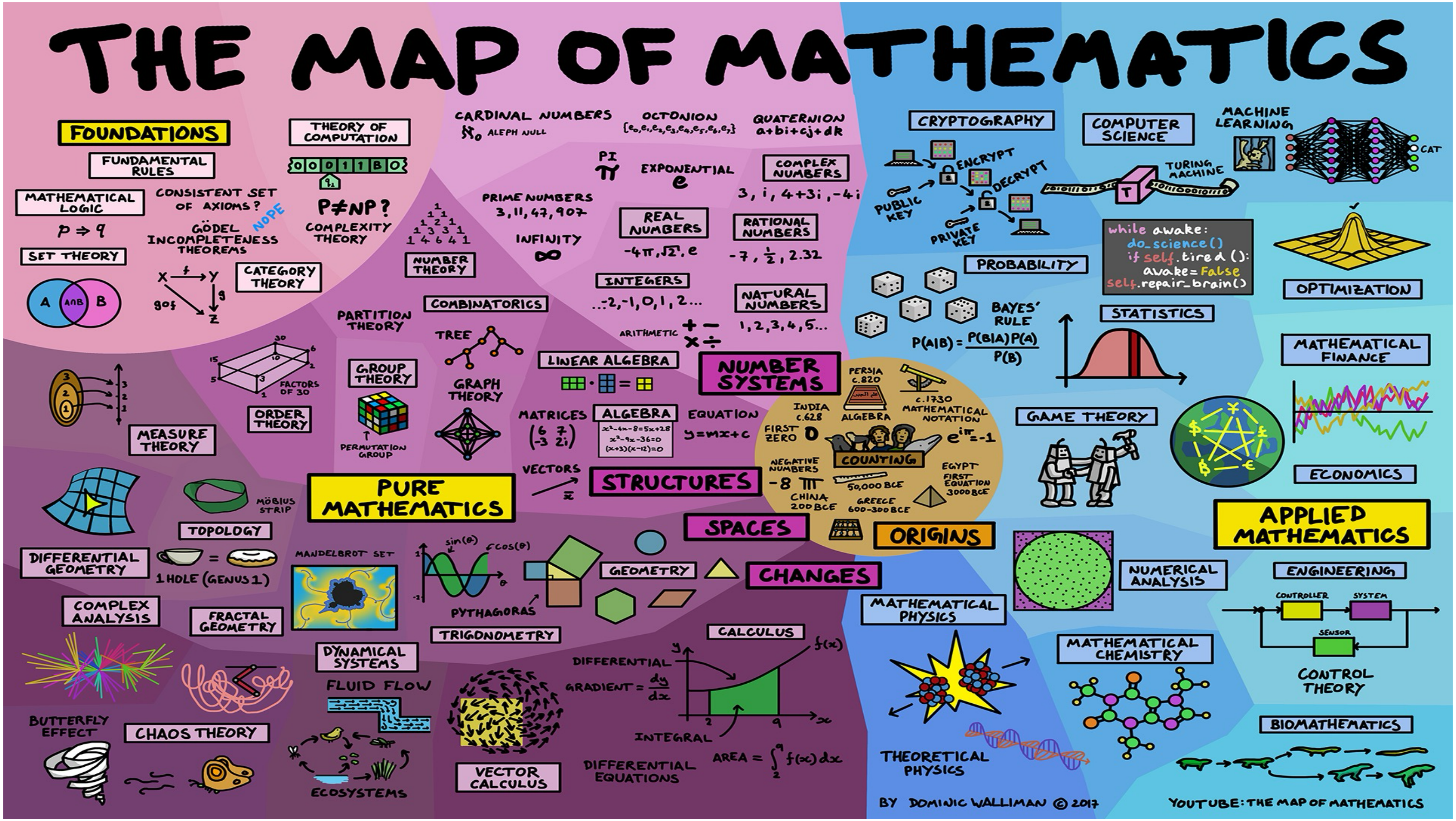


Mathematics, Computer Science, and Education

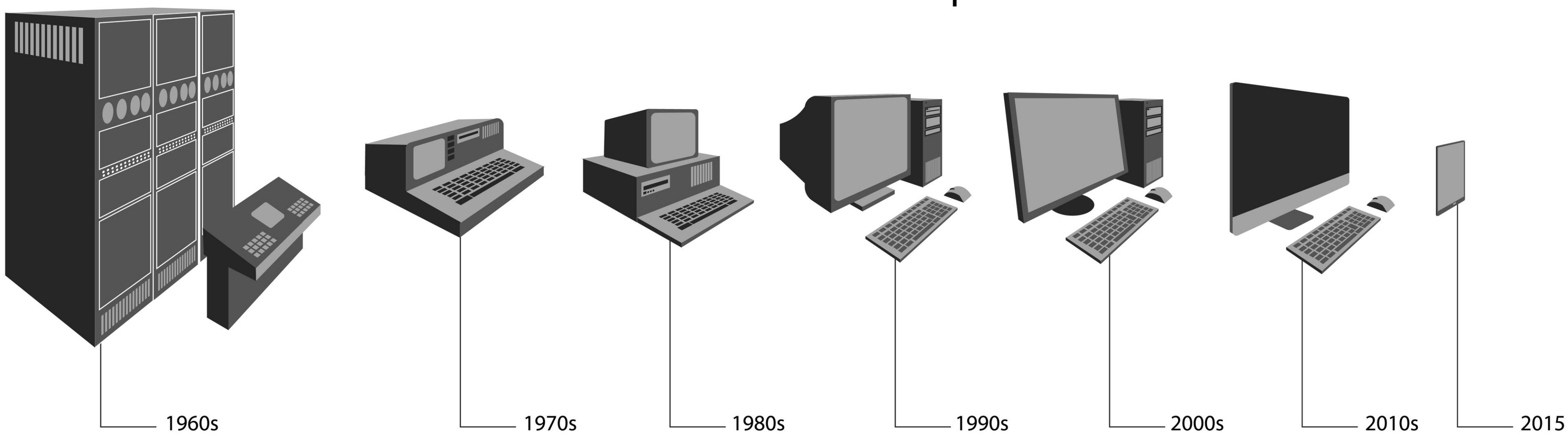
NUI Galway Taster Day


WOODEN ITEMS
WOOD BLOCK
50p
WOOD PLANKS
£1
WOOD SLAB
50p
WOOD STAIRS
£1
ALL WOODEN ITEMS COME IN A VARIETY OF WOOD TYPES INCLUDING:
- OAK
- SPRUCE
- BIRCH
- JUNGLE
- ACACIA
- DARK OAK



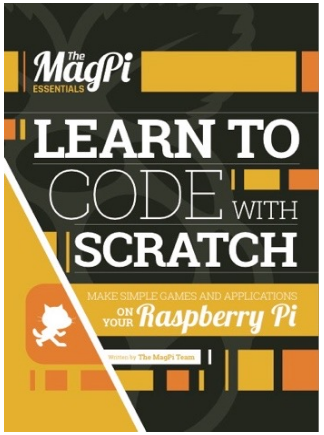

THE MAP OF MATHEMATICS
FOUNDATIONS
FUNDAMENTAL RULES
MATHEMATICAL LOGIC
p ⇒ q
CONSISTENT SET OF AXIOMS?
GÖDEL INCOMPLETENESS THEOREMS
NOPE
SET THEORY
A B A∩B
CATEGORY THEORY
X f Y g g∘f Z
THEORY OF COMPUTATION
P≠NP?
COMPLEXITY THEORY
CARDINAL NUMBERS
ℵ0 ALEPH NULL
OCTONION
QUATERNION
a+bi+cj+dk
PI π
EXPONENTIAL e
COMPLEX NUMBERS
3, i, 4+3i, -4i
PRIME NUMBERS
3, 11, 47, 907
INFINITY ∞
REAL NUMBERS
-4π, √2, e
RATIONAL NUMBERS
-7, 1/2, 2.32
INTEGERS
...-2, -1, 0, 1, 2...
NATURAL NUMBERS
1, 2, 3, 4, 5...
ARITHMETIC + − × ÷
NUMBER THEORY
COMBINATORICS
PARTITION THEORY
TREE
GRAPH THEORY
GROUP THEORY
PERMUTATION GROUP
ORDER THEORY
FACTORS OF 30
MEASURE THEORY
LINEAR ALGEBRA
MATRICES
ALGEBRA
EQUATION
y=mx+c
VECTORS
NUMBER SYSTEMS
STRUCTURES
SPACES
CHANGES
PURE MATHEMATICS
TOPOLOGY
MÖBIUS STRIP
1 HOLE (GENUS 1)
DIFFERENTIAL GEOMETRY
COMPLEX ANALYSIS
FRACTAL GEOMETRY
MANDELBROT SET
sin(θ) cos(θ)
PYTHAGORAS
GEOMETRY
TRIGONOMETRY
DYNAMICAL SYSTEMS
FLUID FLOW
ECOSYSTEMS
BUTTERFLY EFFECT
CHAOS THEORY
VECTOR CALCULUS
CALCULUS
DIFFERENTIAL
GRADIENT = dy/dx
INTEGRAL
AREA = ∫ f(x) dx
DIFFERENTIAL EQUATIONS
ORIGINS
PERSIA c.820
ALGEBRA
c.1730 MATHEMATICAL NOTATION
INDIA c.628
FIRST ZERO 0
COUNTING
NEGATIVE NUMBERS -8
CHINA 200 BCE
50,000 BCE
GREECE 600-300 BCE
EGYPT FIRST EQUATION 3000 BCE
CRYPTOGRAPHY
ENCRYPT
DECRYPT
PUBLIC KEY
PRIVATE KEY
COMPUTER SCIENCE
TURING MACHINE
MACHINE LEARNING
CAT
while awake:
do_science()
if self.tired():
awake = False
self.repair_brain()
OPTIMIZATION
PROBABILITY
BAYES' RULE
P(A|B) = P(B|A)P(A) / P(B)
STATISTICS
MATHEMATICAL FINANCE
GAME THEORY
ECONOMICS
APPLIED MATHEMATICS
NUMERICAL ANALYSIS
ENGINEERING
CONTROLLER
SYSTEM
SENSOR
CONTROL THEORY
MATHEMATICAL PHYSICS
MATHEMATICAL CHEMISTRY
THEORETICAL PHYSICS
BIOMATHEMATICS
BY DOMINIC WALLIMAN © 2017
YOUTUBE: THE MAP OF MATHEMATICS

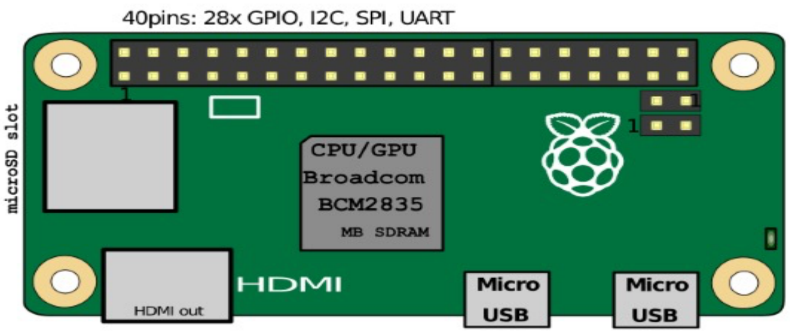
Evolution of computers

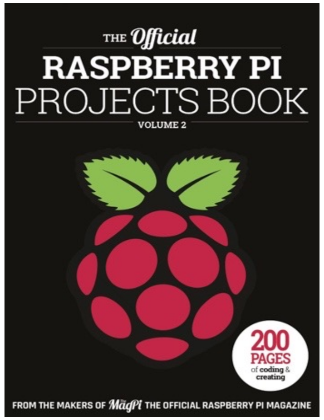


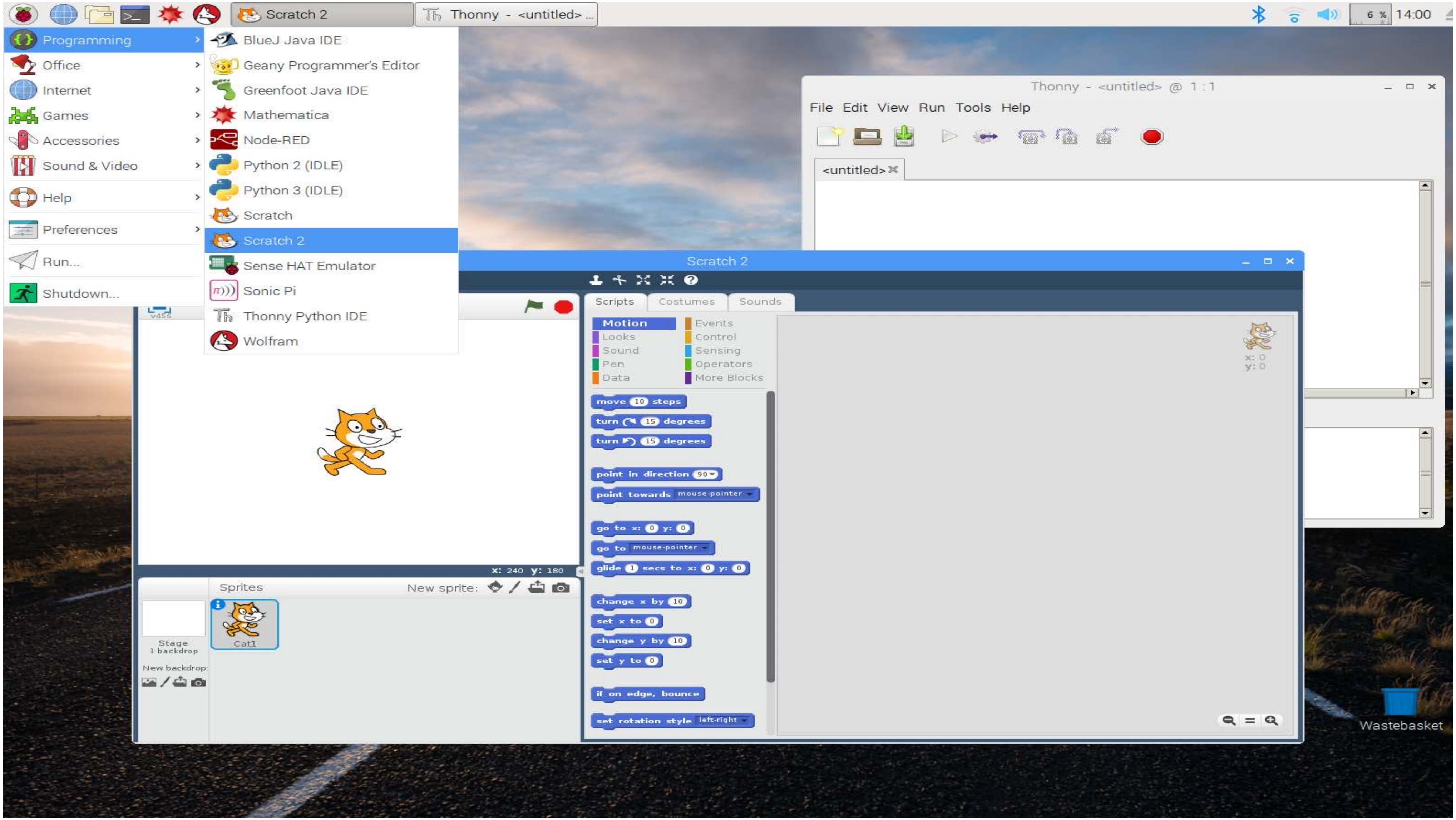
- https://www.youtube.com/watch?v=QvyTEx1wyOY

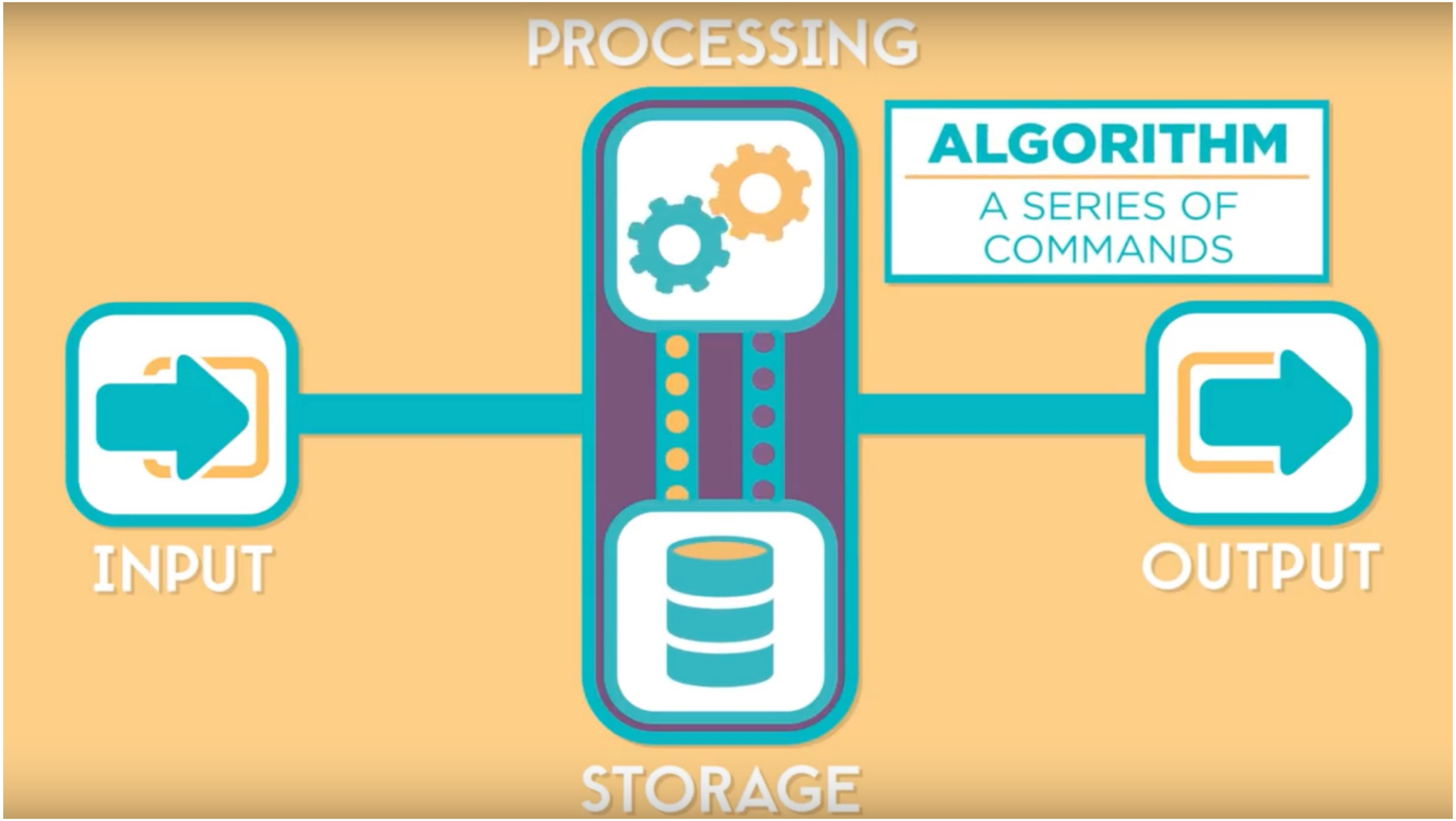
Raspberry Pi®








Scratch 2
Thonny - <untitled> ...
14:00
Programming
Office
Internet
Games
Accessories
Sound & Video
Help
Preferences
Run...
Shutdown...
BlueJ Java IDE
Geany Programmer's Editor
Greenfoot Java IDE
Mathematica
Node-RED
Python 2 (IDLE)
Python 3 (IDLE)
Scratch
Scratch 2
Sense HAT Emulator
Sonic Pi
Thonny Python IDE
Wolfram
Thonny - <untitled> @ 1 : 1
File Edit View Run Tools Help
<untitled>
Scripts
Costumes
Sounds
Motion
Looks
Sound
Pen
Data
Events
Control
Sensing
Operators
More Blocks
move 10 steps
turn 15 degrees
turn 15 degrees
point in direction 90
point towards mouse-pointer
go to x: 0 y: 0
go to mouse-pointer
glide 1 secs to x: 0 y: 0
change x by 10
set x to 0
change y by 10
set y to 0
if on edge, bounce
set rotation style left-right
x: 240 y: 180
Sprites
New sprite:
Stage
1 backdrop
New backdrop:
Cat1
Wastebasket


PROCESSING
ALGORITHM
A SERIES OF COMMANDS
INPUT
OUTPUT
STORAGE

New Words!

Algorithm

Say it with me: Al-go-ri-thm

A list of steps that you can follow to finish a task

Program

Say it with me: Pro-gram

An algorithm that has been coded into something that can be run by a machine

Programmer \\ Hacker \\ Software Engineer \\ Coder \\ Computer Scientist \\ Developer

The Math Process

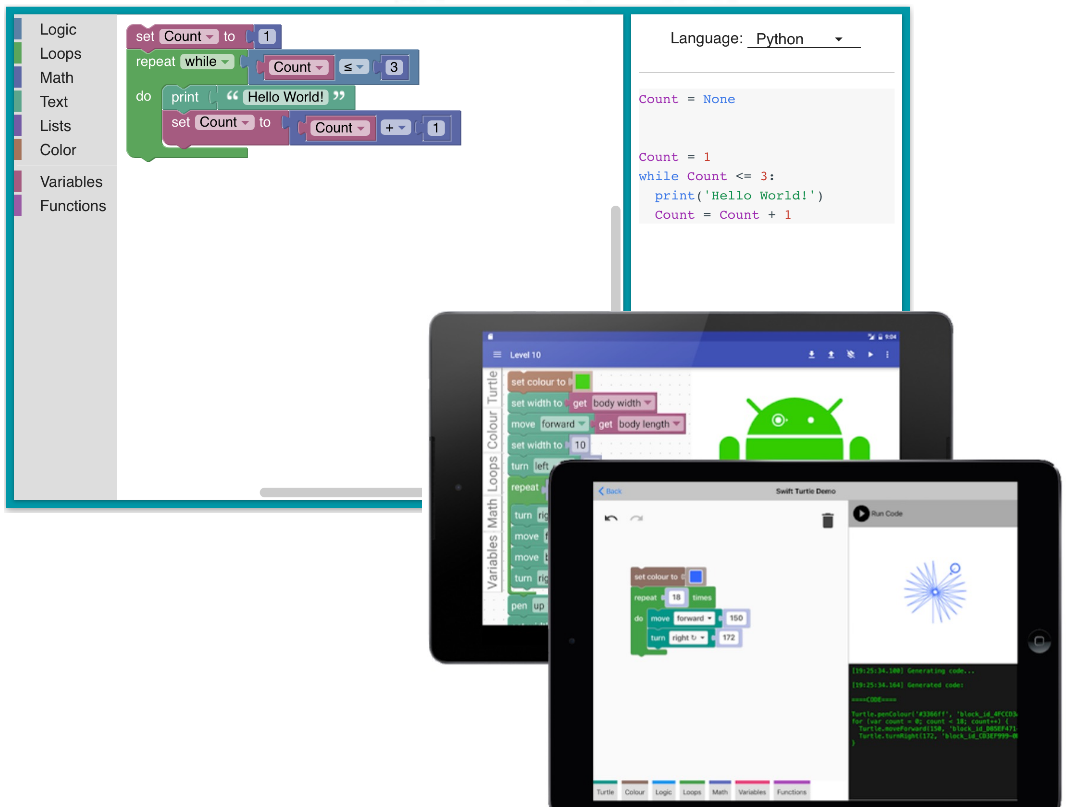
1. Math teaches understanding and communication through abstract language.

2. Math teaches how to work with algorithms.

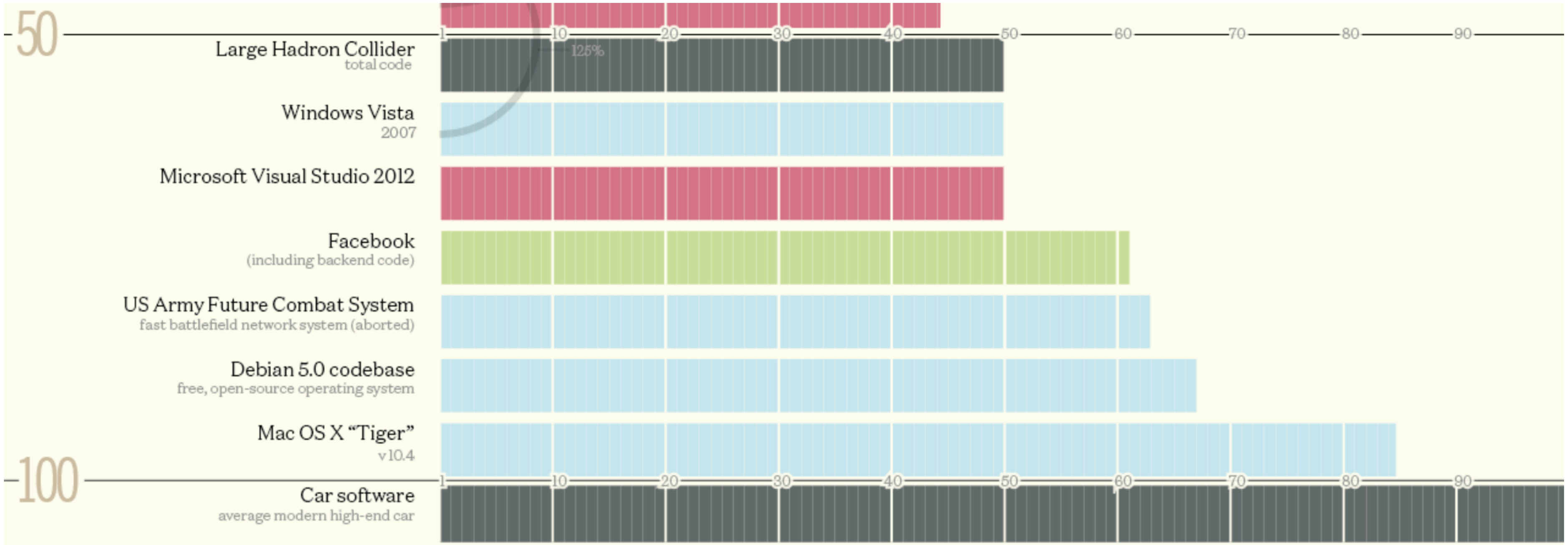
3. Math teaches students how to analyze their work.

SCRATCH

Blockly
Google
A Visual Programming Language



How Many Millions of Lines of Code Does It Take?



C O
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Grades 6-12

Courses by Code.org

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Learn the fundamentals of Computer Science with drag & drop blocks. Create your own drawings and games.

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App Lab

Start with App Lab: an introductory programming environment where you can design an app, code with blocks or JavaScript to make it work, then share your app in seconds.

Learn more

Game Lab

Ready to go further? Game Lab is a more complex programming environment where you can make animations and games with characters that run, jump, fly and more.

Learn more

Web Lab

Web Lab is a programming environment where you can make simple web pages using HTML and CSS. Design your web pages and share your site in seconds.

Learn more

Name: Cian Mac Liatháin **ID:** 11476078

ED305 Assignment 1: Inclusive Teaching Paper

A. Context:

This lesson plan is designed based on the context of the school I undertook teaching practice in this year. The school is mixed-sex, in an urban setting, and caters for roughly 300 post-primary students. The school is classified as a DEIS school, and the student population consists of diverse socio-economic and cultural backgrounds, including minority ethnic backgrounds as well as members of the Travelling community. Some students have had difficulty in other schools previously and have now found a place in this school which prides itself in providing an education for all types of students based on their individual strengths, emphasising an interdenominational, academic and vocational ethos. The school also provides education in the form of post-Leaving Cert PLC courses and part-time adult evening courses, which are beneficial to all types of students as alternative pathways to higher education.

The class group is a Transition Year maths class of 20 students. As a Transition Year group, they are a mixture of both higher level and ordinary level maths students. The students are primarily from low socio-economic backgrounds and are culturally diverse. The group includes 5 visiting Spanish students who have English as an Additional Language, as well as 2 non-Traveller minority ethnic students.

This lesson will be their first lesson in Statistics since completing the Junior Certificate the year before. For the students visiting from Spain, they have previously visited the same statistical topics during their time studying for the Spanish equivalent of the Junior Certificate, *Educación Secundaria Obligatoria.* All students have previously encountered the ideas of data collection and statistical analysis, and in this lesson they will be refreshing their knowledge by carrying out their own statistical investigation. This will provide them with a solid statistical foundation before carrying out further statistical investigations in later classes.

Nearly all students in this class exhibit poor engagement with traditional methods of teaching mathematics through repeated practice of questions and rote learning. Their disengagement leads to a lack of communication in the class, as well as difficult to control behaviour. Furthermore, the visiting students from Spain only talk to each other, if at all. Similarly, the two ethnic-minority students only sit next to and interact with each other. The visiting students frequently refuse to answer questions asked of them, simply stating that they 'can't do it'. None of the students are expected to have a textbook, and some normally don't have a pen or a copy, due to difficulties affording them. In previous classes, two female students expressed a dislike of mathematics and dismissed any discussion about its usefulness, commenting "I can't go to college anyway, I'll only be a hairdresser".

In my opinion, inclusive teaching in the context of this class involves making the topic of mathematics more accessible to all students, in an effort to promote communication and participation in the class. My own teaching and communication should be done in a way that is conscious of students with English as an additional language. The lesson should be designed in a way that promotes peer learning and communication between all types of students, as well as facilitates students that aren't coming into the class with their own materials. I hope to encourage students towards mathematics by highlighting its relevance to all professions and dismissing myths about 'not being a maths person'. My aim is to ensure that neither language, culture nor gender is a barrier to the mathematical learning outcomes.

B. Inclusive Lesson Plan:

Highlighting the diversity of students' personalities, experiences, and viewpoints, Tomlinson (2017) emphasises the importance of teachers carrying out differentiated instruction in their classrooms. A differentiated classroom "provides different avenues to acquiring content, to processing or making sense of ideas, and to developing products so that each student can learn effectively" (Tomlinson, 2017, p.1). In order for me to create an inclusive lesson, I feel it is important to carefully consider not only the design of the lesson itself but also the principles that underly my teaching. Tomlinson (2017) outlines principles that "can be helpful in ensuring that struggling learners maximise their capacity in school", and from these I have identified those that are important to me as a pre-service teacher:

Teach with a growth mindset and with a belief of the capacity of each child – this is a principle that I feel is very important, as it avoids placing blame at the feet of the students for difficulties in learning. Believing that all the students in my class have an inherent capacity for growth and learning allows me to focus on improving my own teaching to meet their needs. For example, when the EAL students sometimes declare their inability to do any maths, I won't simply agree with them and focus on other 'easier' students. By possessing a belief in their capacity for growth and learning, their feelings of inability give me valuable insight, and motivate me to identify challenges from their point of view. Perhaps my English speaking is too fast, or my Irish accent is too strong. These are easily fixable problems as long as I remain conscious of them, which I hope to do as I carry out the lesson outlined below. Tomlinson (2017, p.27) advises that working to ensure EAL students feel acknowledged and accepted can greatly improve their learning and participation in class, and I hope to do this by consistently offering opportunities to talk, as well as integrating Spanish key vocabulary lists into the lesson to "build bridges between students' first languages and English".

Pay attention to relevance – this is an important principle that I feel will help all types of students in the class. In different ways, many students fail to see the relevance of any maths topic approached in a traditional way that doesn't acknowledge the diversity of the students in the class. Students who don't think they have any mathematical potential grow disengages in a similar way to the EAL students, after too much abstract jargon is used in my teaching. This lesson is designed in a way that emphasises the relevance to the students' lives, and will hopefully work as a foundation in making future mathematics lessons accessible as well.

Learning Outcomes

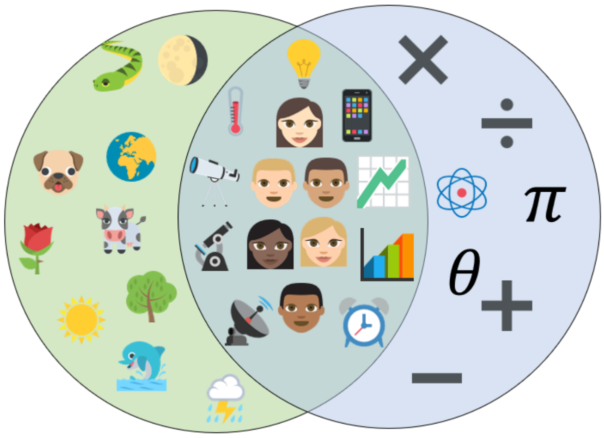
One of the main learning outcomes for this lesson is for the students to develop a statistical view of the world and grow more statistically aware. I hope for students to develop an understanding of the importance and relevance of mathematical thinking to everyday life.

By the end of the lesson, students should be able to understand what is meant by the Data Handling Cycle. When explained with worldwide examples, I expect that students will be able to recognise statistical investigations, data collection and its applications already present in their everyday lives.

Carrying out their own statistical investigation, students will work in pairs to practice and know how to mathematically calculate the Mean, Median and Mode of data they collect, and through discussion with other pair-groups discuss their usefulness.

Lesson Layout

The lesson will be opened with a PowerPoint slide of a Venn diagram, with one side containing images/symbols related to nature, another with symbols related to mathematics and the centre with symbols of humans and examples of how we apply the right side to the left. This will be used as a hook, to introduce mathematical, statistical and scientific thinking in the context of our shared daily experiences. Example of this given below:



The use of open questioning has been described by Small (2012, p.6) of Mathematics as a "core strategy" of differentiating instruction in a mathematics classroom. Open questioning allows for teachers to encourage the expression of different points of view and promotes confidence in students (Small, 2012) This will be used to prompt discussion in the class about the students' opinions of mathematics and its applications. Example questions include: "Do you think maths is useful? What are uses of maths? Has maths changed our lives? How does maths make you feel?". The teacher can describe the centre of the Venn diagram by giving examples of how we use maths to make sense of the world, how it is prevalent in our everyday lives, and giving examples of how different professions use maths. Maths is framed as a universal language.

After engaging with the students, the lesson continue on with the concept of a Venn diagram, asking "Why do we say Pythagoras' Theorem? Why is this called a Venn diagram?". The NCCA (2006, p.70) has identified that exploring "the contribution of diverse cultures to our mathematical culture" is a key aspect of an inclusive Mathematics programme. The teacher will describe mathematics as a joint effort, featuring contributions from many different types of people and dispelling the myth of a singular type of 'maths person'. The teacher will show the photo of, and give brief descriptions of a variety of people who have contribute to the field of mathematics. For example:

John Venn – a male English mathemetician famous for contributions to statistics, including the Venn diagram.

Hypatia – a famous female mathematician from Alexandria, Egypt, c.350-370 AD. Also a philosopher and astronomer.

Yutaka Taniyama – a male Chinese mathematician significant contributions to modern geometry that we use today.

Maria Assumpcio Cata i Poch – a female Spanish mathematician contributions to mathematical analysis, important work involved in calculating orbits.

David Blackwell – a male African-American mathematician significant contributions to statistics and probability.

Florence Nightingale – a female English mathematician who made significant improvements to data collection and using it to improve health services.

The lesson will continue by asking students about the use of data in modern society. As all students have smartphones, the concept will be introduced in terms of asking how companies like Apple, Google, Amazon, Facebook collect and use data. Pictures of two different Amazon reviews, in Spanish and English, will be used to discuss quantitative and qualitative data, their uses, and the accuracy of data and statistics (e.g. an item with a single wordless 5-star rating vs an item with thirty 4.5-star reviews with detailed descriptions).

According to the NCCA (2006, p.69), presenting opportunities for students to "examine information on local and global issues" is an important aspect of an inclusive Mathematics programme in terms of intercultural education. The teacher will present the interactive COVID-19 map organized by John Hopkins University (2020). This will be used as an example of the shared community of mathematics, asking questions like: " Why should we be concerned about global data? How does this data from other countries help us here in Ireland? Is it important for countries to communicate with each other?". This resource is particularly useful for prompting discussion on statistical accuracy and data collection, as there are interactive statistics available that allow the teacher to discuss why infection rates could be lower in countries that are simply testing less, for example.

Another important aspect of an inclusive Mathematics programme, according to the NCCA (2006, p.69) is giving students opportunities to "engage in group activities and investigative learning". Tomlinson (2017) advises that there are different requirements for effective group work to take place. The goals of the task should be made clear to the students, and groups should be designed based on "group members' strengths and interests" as well as in a way that "requires group collaboration to achieve shared understanding and quality outcomes (Tomlinson, 2017, p.48). Based on this, the members of each group-pair will be chosen by the teacher in order to promote the collaboration and communication of the students between different class subgroups. This investigation is designed in a way that promotes communication between the two members of each group in addition to communication between the groups themselves. This is guided by principles of co-operative learning (Clarke, 1994)

The students will investigate the contents of Smarties tubes provided to each pair of students by the teacher, guided by the worksheet attached in the Appendix below. The teacher will also provide colouring pencils and pens to students without any. The investigation is based around the fact that Smarties tubes don't give specific numbers for their contents, instead using an approximation per tube. The students will investigate, counting the Smarties, calculating the Mean/Median/Mode and graphing the data. After the initial investigation carried out by pairs of students, one member of each pair will travel to students that investigated in other pairs. The class as a whole will compare their data to each other and will try to get a whole-class perspective.

The investigation is intended to promote discussion between all students of the class including EAL students and ethnic-minority students. From the beginning of the lesson the teacher will have been contributing to a bilingual word key, and students are encouraged to communicate in any way they feel comfortable. As it is being filled, the teacher will attempt to frame the foreign language and other cultures in a positive light. For example, it could be explained that many languages are closely related and all languages are useful. An example discussion point could be, which word has a more intuitive meaning, 'Accuracy' or 'La exactidud'? Example word key:

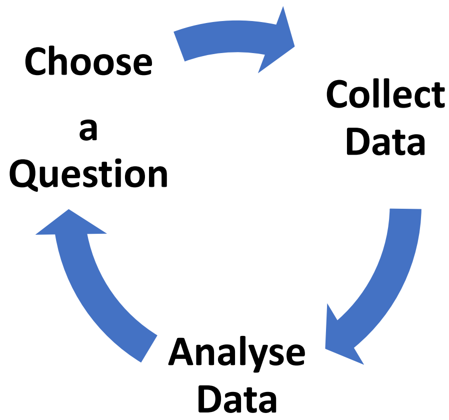
English	**Spanish**
Data	Datos
Collect	Recoger
Analyse	Analizar
Interpret	Interpretar
Investigation	Investigación
Statistics	La Estadística
Accuracy	La exactitud
Qualitative	Cuantitativa
Quantitative	Cualitativa
Graphs	Gráfico
Bar chart	Gráfico de barras
Measure	Medida
Mean	Media
Median	Mediana
Mode	Moda
Range	Rango

After the students carry out their investigation, communicating in pair-groups and between pair-groups, the teacher will congratulate them on carrying out a successful statistical investigation. An emphasis will be placed on the importance of collaboration, and the teacher will explain how future lessons will continue to build on their statistical reasoning and co-operation. A few examples of Census@School (2020) data and statistics will be shown from international classrooms, and it will be explained that we will carry out similar investigations in the coming classes.

Smarties Investigation

Try not to open them until Part 2! 😉

Aim: To develop our statistical knowledge through investigation, using the Data Handling Cycle:



Part 1 - Choose a Question:

- Today we'll investigate the contents of a tube of Smarties. Can we find any information on the tube telling us *how many Smarties are inside?*

___

- Based on the picture on the tube, *how many different colours should there be?*

___

- Can you come up with a question that describes what we want to investigate?

___

Part 2 - Collect Data:

- Pour the Smarties out and find the Total Amount: ____________
- Sort them in columns of each colour.
- Gather data for each colour:

Colour										
Amount										

Part 3 - Analyse Data:

- How many different colours are there? ______________________
- What is the **Mean** amount of Smarties?
- What is the **Median** amount?
- What is the **Mode?**
- Calculate the **Range** of the amounts of Smarties:
- Are these the same as the results of the person next to you?
- Draw a **Bar chart** of your Data**:**

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ED305 Assignment 2: Using Psychology and Sociology of Education to Make Sense of My School Experience

In Issue One I will explore how my sense of self-concept and self-efficacy have influenced my view of inclusive education. In Issue Two, I will try to understand better how the idea of meritocracy effected my school experience with a working-class student.

These issues took place during this year's teaching practice that I undertook in an urban DEIS school that facilitates 300 students. These students are of diverse cultural backgrounds, and predominantly low socio-economic backgrounds. My placement in the school involved: teaching a mainstream mixed-level Transition Year maths class, observing and team-teaching a First Year ordinary level maths class, resource hours with a Leaving Cert maths student during school, resource hours as part of the After School Study programme provided to the students for free by the school.

Issue One: Identity

This 2019/2020 academic year marked my return to college, after multiple failed attempts at progression. During those previous failed attempts, I was diagnosed with a mental health condition, and this greatly affected my beliefs about myself as a student as well as a student-teacher. Before this teaching practice began, I already felt anxious and defeated. These emotions were prominent throughout the beginning of my teaching practice. As my teaching practice continued, however, my emotions and outlook improved greatly, and this was reflected in my teaching and then in the behaviour and learning of the students I had the pleasure of teaching.

I chose this experience as I feel it's important to better understand the relationship between my identity as a teacher and my performance in class. A deeper understanding of how mental disability, academic performance and teaching interact will also help me make informed decisions in terms of inclusive education. It has been identified that in an inclusive education system, "every child and young person deserves an equal chance to access, participate in and benefit from education" and "each person should have the opportunity to reach her/his full educational potential" (DES, 2005, p.7). Students with special educational needs should be educated alongside their peers in mainstream classes as much as possible (EPSEN Act, 2004), and an important factor in the success of inclusive education is how teachers themselves perceive disability and special educational needs (Butler & Shelvin, 2001). Based on this, I will investigate my own experience with disability and how it has influenced my approach to inclusive teaching.

I was relatively confident Education in my academic and teaching potential when I first began this Bachelor of Mathematics & Education. Over time, however, due to experiencing traumatic events and a resulting diagnosis of Complex Post-Traumatic Stress Disorder, my identity as a growing teacher and as a hard-working student seemed to disappear and I found myself incapable of submitting written assignments for 5 years. Students experiencing post-traumatic stress have been shown to be at an increased risk of school burnout and developing depressive symptoms (Lucier-Greer et al., 2019). Learning this has helped me make sense of how the excitement and pride I normally felt while teaching was replaced by feelings of inability and emotional exhaustion. My teaching performance began to reflect my negative emotions, with a negative classroom atmosphere characterised by student frustration and behavioural and motivational issues. These issues and my teaching performance all improved significantly by the time my teaching practice ended. I believe this was due to changes in my self-concept.

Self-concept has been identified as an important factor in teacher development, having been shown to influence approaches to teaching, beliefs about students and risk of teacher burnout (Guskey, 1988; Friedman & Farber, 1992; Woolfolk et al., 2013). Described as "a multidimensional structure of identity which includes self-esteem, group identity, and self-efficacy" (Bogart, 2014, p. 108), self-concept is also something that can be strongly affected by the acquisition of a disability, something that often causes "a stunning blow to one's identity" (Smart, 2019, p.361). Being able to make sense of self-concept and identity should help inform my future teaching in order to better support students who are members of minority groups, such as those with disabilities.

Social Identity Theory asserts that individuals strive for positive self-concept, by identifying and relating to positive aspects of social groups they feel a part of (Tajfel & Turner, 1979). People who are members of stigmatized groups are at risk of negative self-concept, as they take part in a society where the dominant group may view the stigmatized group in a negative light (Crocker & Major, 1989). To improve their self-concept and manage stigma, members of stigmatized groups may attempt to assimilate, distancing themselves from their minority group and ultimately taking on the majority group's values and negative view of the minority group they belong to (Tajfel and Turner, 1979; Dunn & Burcaw, 2013). Those with "invisible disabilities such as learning disabilities or mental illness" (Bogart, 2014, p.113) are at higher risk of negative self-concept as they are less likely to find comfort in similar others due to their stigma being concealable (Frable et al., 1998) and are also more likely to be misniterpreted if their disability is undisclosed. This insight has helped me recognise a negative correlation between my understanding of my disability and my identity as a teacher. The more I improved at understanding and managing my disability, the less capable I felt as a teacher. New limitations on my ability to teach caused me to feel distant and different from éxperienced teachers who had once inspired me, and left me wondering if I was doing a disservice to the students and their future.

Alternatively to developing negative self-concept, people with disabilities may begin to develop a sense of disability pride by identifying as part of their minority group and reinterpreting the stigmatization in a positive light (Dunn and Burcaw, 2013). This results in a more positive self-concept, and it is thankfully something that I have identified as happening by the end of my teaching placement as a result of increasing self-efficacy.

Bandura (1997, p.3) defines self-efficacy as "beliefs in one's capabilities to organise and execute the courses of action required to produce given attainments". My teacher self-efficacy, for example, can be described as my beliefs about my own capacity "to bring about desired outcomes of student engagement and learning, even among students who may be difficult or unmotivated" (Tschannen-Moran & Hoy, 2001, p.783). Evidence suggests that significant changes to self-concept and self-efficacy in one context can generalize to other situations (McConnell, 2011; Bandura, 1977). Furthermore, it has been shown that teachers require a clear sense of self in order to respect and bring about cultural or academic change in their students (Warin et al., 2006). For these reasons, I believe that my academic, teaching and disability self-concepts are closely related and strongly dependent on my sense of self-efficacy. I hope to better understand how changes in my self-efficacy throughout teaching practice have made these self-concepts and my teaching more positive.

Bandura (1977, p.195) advises "expectations of personal efficacy are based on four major sources of information": physiological states, vicarious experience, verbal persuasion, and performance accomplishments.

Beginning with physiological states, I can now identify an important interaction I had with my co-operating teacher on my first day of teaching the Transition Year class. Frozen with anxiety in the days prior, my sense of self-efficacy was extremely low. The morning of the class I asked the teacher could I observe the teacher instead of teaching, due to strong feelings of anxiety and not feeling capable of doing it. The teacher kindly refused and advised the only way to improve that was to teach through it. My teaching in the following few weeks was hindered by this physiological state, I dismissed modern teaching methods as being out of my ability and was pre-occupied with what could go wrong in the class, managing behaviour through traditional teaching methods. A notable memory I have with them was after a long class of controlling classroom behaviour, one student commented "looks like you're having a bad day today, sir". This prompted a strong realisation that my emotions were not only affecting my teaching style but could also be perceived by the students and influence their behaviour. I was reminded that they were human too and, like me, would benefit from a positive atmosphere. I began to try making the classes as relatable and enjoyable as possible for the students. Over time my own anxiety, fear and negativity diminished, and self-efficacy improved. By bringing a more relaxed and positive mindset to the class the students began to engage more and felt more relaxed and positive towards the maths too.

In terms of vicarious experiences, my co-operating teacher was happy to let me observe her other classes. I took the opportunity to observe as many classes as I could and was surprised that she also had difficulty with some students' behaviour. This helped me identify an aspect of my own teaching I had been struggling with. Occasionally I would be completely ignored by certain students. Situations like that would leave me feeling very self-conscious about my teaching ability. I had begun to conclude that I just wasn't good enough to teach anymore. Through observing my co-operating teacher, I could see that she was ignored too, and had to deal with far wilder classroom behaviour than I had. A key difference that I observed was that she kept persisting and wasn't embarrassed or afraid to repeatedly pursue the outcomes that she wanted from the students. I began to do the same in my class to great results, students that had previously ignored me now began to focus on the maths and engaged in the classroom activities. This experience sparked a change in my self-efficacy in terms of how I felt my disability effected my potential growth. I had previously seen difficulties in the classroom as being a fixed result of my disability, rather than in the context of me being a pre-service teacher. After observing, relating to, and learning from a more experienced teacher I now felt hopeful that I still had the capacity to grow as both a teacher and person.

Verbal persuasion is something that stands out in improving my sense of self-efficacy. Closer to the beginning of the year, a negative academic self-concept was strongly influencing my self-efficacy as a teacher. If I was someone struggling to complete assignments, who was I to ask students to challenge themselves academically? An interaction with a lecturer left a lasting impression, where I spoke to them about my mental health condition. I was surprised that they saw me as a hard worker despite previous failings, and that they suggested my experience would better prepare me to teach disadvantaged students. This description of myself didn't fit with my negative self-concept so was initially dismissed, but I tried to consider the possibility that it was true. Following this interaction, I had an experience with a First-Year student who emphatically stated, "I can't do that sir, I can't read!". My initial reaction was a feeling of low self-efficacy, waiting for the Special Needs Assistant to help, as

I felt unprepared and unsure if I knew how best to help the student. Remembering my lecturer's comments, however, I noticed how strongly I related to how this student was feeling. I felt motivated to help them through the challenge of reading the maths question in a way that I would want to be if I was struggling with it: by explaining that I can find reading hard sometimes too, leading by example, keeping a positive attitude, and praising them when they progressed through it. The student grew enthusiastic after getting the maths questions right after successfully reading it, and we continued through the rest of them together. At the end of the class the SNA commented that she saw I was doing good work with the student. This comment cemented the increased self-efficacy that I felt after putting my lecturer's advice into action.

Lastly, I've been able to identify how performance accomplishments have informed my sense of self-efficacy also. As my teaching practice progressed, and as the other examples above improved my sense of self-efficacy, I began to gain confidence in planning my lessons away from traditionalist values and more towards those of social constructivism inclusion. I found it challenging to carry out lessons designed around: activities and group selection scaffolded to maximises the learning of all students, relatable examples that promote student discussion and their natural curiosity for learning. My sense of self-efficacy improved weekly based on positive experiences with the class and seeing their learning improve. These efforts culminated in a teaching inspection that praised me for my lesson preparation, teaching style and my rapport with the students, which is very much in contrast with what I felt was possible at the beginning of the teaching placement.

Sociologically, I have shown how social identity has had an impact on my understanding of minority groups and my ability to carry out an inclusive teaching philosophy. By applying the psychology of education, I have identified examples of how improvements in self-efficacy have brought about positive changes in my academic, teaching and disability self-concepts. Feelings of inability and defeat have transformed into a sense of pride and excitement for my future as a teacher with a disability. I have gained important insight into how self-efficacy and self-concept can affect academic performance, as well as how they can be improved, and hope to use this insight in future to create learning environments that promote self-efficacy and to help maximise my students' achievement.

Issue Two: Meritocracy

During my teaching practice I was assigned a resource class with a working-class Leaving Cert student, described to me as a "weak" and a "problem student". I was advised that I shouldn't expect them to show up for the class. After two weeks, I found the student and convinced them to attend the resource class. Apprehensive at first, he spoke to me about his dislike for mathematics and teachers in general, and that he wasn't sure if he was going to sit the Leaving Cert at all. He struggled with any topic I'd begin to teach, but after focusing on more foundational mathematical concepts the student started growing confident and started doing much better with Leaving Certificate questions. The student spoke to me about how he liked coming to my class because I treated him differently than other teachers and didn't "give out to him" for his behaviour. The student showed a new willingness to learn and asked for extra help after school outside of my scheduled hours, as he couldn't afford private tutoring. I helped him as much as I could, and throughout the year his performance, behaviour and outlook on education improved greatly.

I chose this experience because it made me feel conflicted about my position in the education system. I felt morally obliged to seek out this student and try to help them as much as I could, but as I was doing so, part of me wondered if it was due to some pre-service teacher naivety. If I was a full-time teacher would this extra help be fair to other students? Would I eventually start seeing certain students as "weak" and lower my expectations? I hope to make more sense of this experience through research into the idea of meritocracy.

Meritocracy has been described as a society where the talents, attitude, and hard work of an individual is what determines their success or failure in society (Considine and Dukelow, 2009). Originally coined in a satirical work by Michael Young (1958), there is widespread allegiance to a meritocratic viewpoint in Ireland (Kennedy and Power, 2010). With merit being so concisely defined as "Intelligence + Effort = Merit" (Young, 1958, p.94), I feel like a deeper understanding of intelligence and effort in the case of this student will help me make sense of this experience.

The first psychologist to investigate intelligence scientifically believed that there were two factors involved in cognitive performance, a general intelligence factor and situation specific factors (Spearman, 1904). For example, remembering a set of numbers would involve the general *g* factor as well as the specific *s* factor related to recall something from memory. This *g* factor effects performance on all intellectual tasks, and Spearman suggested that "*g* alone is of psychological significance (Roberts & Lipnevich, 2012, p.9). The concept of general intelligence has gone on to inform many other theories of intelligence (Cattell, 1963; Carroll, 1993). Cattell (1963), for example, divides general intelligence into fluid and crystallized intelligence. Fluid intelligence is a relatively fixed aspect of intelligence that is tied to brain development and relates to an individual's "mental efficiency [and] non-verbal abilities" (Woolfolk et al., 2013, p.128). Crystallised intelligence includes skills learned throughout life and is developed "by investing fluid intelligence in solving problems" (Woolfolk et al., 2013, p.129). The most widely accepted theory combines the work of Cattell with other psychologists, 'Cattell-Horn-Carroll' theory proposes a hierarchical three-level model of intelligence (Sternberg, 2012). General intelligence, at the top level, informs a second level of

8 broad abilities including fluid and crystallised intelligence, and visual-spatial reasoning. These broad abilities connect to over 80 narrow abilities at the third level, like writing ability and listening ability. Dismissing the importance of a general *g* factor, a contrasting view on intelligence is Gardner's (1983) theory of multiple intelligences which views intelligence as 8 distinct abilities, including logical-mathematical, interpersonal and intrapersonal intelligences. Although lacking in scientific evidence, the multiple intelligences view is popular with teachers worldwide (Waterhouse, 2006; Jonsson et al., 2012).

Knowledge of scientific theories of intelligence can be beneficial to educators as they can help us plan lessons in a way that enhance student learning (Lynch & Warner, 2012), and teacher's opinion on intelligence has been shown to have a large impact on student achievement (Jonsson et al., 2012; Dweck, 1999; Carr & Dweck, 2011). This experience initially stood out to me as I felt very uncomfortable when the student was so matter-of-factly described as "weak". This isn't the first time I've encountered this kind of language in school, and I found it disempowering to me as a teacher and insulting to the student. Any time that I have heard descriptions like "weak" it has never inspired hope that I can make a difference as a teacher. The idea that intelligence is fixed and unchangeable, described as entity theory (Dweck 1999), has been shown to decrease motivation, learning and achievement in students (Dweck, 1999; Carr & Dweck, 2011) and is more common in mathematics than other subjects (Rattan et al., 2012). In contrast, teacher's holding an incremental view where intelligence is believed to be changeable over time has been shown to improve those outcomes. The views held by teachers are so influential that students are seen to take them on board themselves, for better or for worse (Rosenthal & Jacobsen, 1992). This knowledge of scientific and implicit theories of intelligence and their potential effects has helped me no longer feel naive or out of place as a pre-service teacher for having my own opinion and practices. With this understanding, I can see now that an important factor in the student's improvement in maths was my own incremental view of intelligence and my attempt to approach his learning from a new angle. I chose to use an iPad game called "Dragonbox: Algebra 12+", which is a very interactive and vibrant game based around the rules of algebra. I was trying to promote learning in a student who was struggling with traditional pen and paper methods, making sure to let him know that I thought he was "very capable", and it was a great success. The student was excited to play the game, and even more excited when we later practiced Leaving Certificate algebra questions and he saw the connections to the rules of the game he was playing. He discussed with me how he didn't initially believe me when I said he could do it, but now was surprised at his progress.

Gardner (1995, p.208), has commented previously that a core aspect of multiple intelligence theory is wanting to take "human differences seriously". Combining this view with an incremental view of intelligence, this experience has shown me the importance of trying to promote learning in all students, by adapting my own teaching rather than writing them off as weak and expecting them to adapt for me. In terms of the meritocratic view, the fact that student achievement can be so dependent on the teacher's own opinions and expectations of them highlights problems of equality in a meritocratic system.

In their investigation of social class barriers in education, Lynch & O'Riordan (1998, p. 445) identify structuralist functionalism as one of the most enduring theoretical frameworks which can be used to explain inequalities in the education system. Functionalist sociology describes a functioning society as consisting of interdependent systems that each contribute to the stability of society, based on a consensus of shared values (Sadovnik, 2011). Durkheim (1956, p.28) viewed education as "the influence exercised by adult generations on those that are not yet ready for social life", socialising students based on the values deemed appropriate by society, and sorting and selecting students based on their abilities. Schools are seen to have different functions that work toward the consensus of a functionalist society: intellectual, political, social and economic (Bennett & LeCompte, 1990).

These different functions have helped me make sense of my interaction with the student. When I first met him, he brought attention to how he avoided participating in maths class because of clashing with his teacher. An example of the social function of education, his teacher had issues with how they spoke and the kind of questions they were asking. After being chastised for his behaviour the student disengaged from learning maths entirely, falling behind his peers with regards to the intellectual function of mathematical understanding. His disengagement, in my opinion, is an example of conflict between functions of the functionalist framework rather than a lack of effort on his part. On the one hand, intellectual functions of education involve students developing an inquisitive mindset to explore, understand and experiment within specific topics chosen by a teacher. Yet social functions require "obedience to authority and 'correct' behaviour" sometimes without question (Bartlett and Burton, 2016, p.17). Both functionalists and Marxists acknowledge the effect of a hidden curriculum, where social inequalities are reproduced due to students receiving differing educations based on social class norms (Lynch, 1989). Functionalists have described economic and social inequality as an inevitable and necessary function of society, to ensure that the most important positions in society will be filled by the most talented people based on shared values (Davis and Moore, 1945). In contrast to this view, conflict theorists like Marxists identify that education promotes the dominant ideologies of the higher class and its structure reproduces the class system through its conflict with the working class (Sadovnik, 2011). This student's situation is an example, to me, of inequality in the education system that shows how different emphasis can be put on the different functions of education depending on a teacher's impression of the student. Some students might be given more freedom to express themselves while others have an educational experience like this student, more focused on modifying their behaviour.

In my opinion, the student's teacher had a very meritocratic view of the education system and of the student themselves. My experience with the student was very different than what I expected based on their teacher's description of a "weak", "problem student". After finally meeting the student and building a rapport with them, I found that I could relate to the student's point of view and saw more fault in the different functions of the education system than in him personally. While his behaviour was sometimes 'incorrect' and off-topic (e.g. calling school 'rubbish', asking for my first name) I found this a great opportunity to connect with him. After time spent in a positive one-on-one environment, the student felt more comfortable and not only applied themselves more to challenging maths questions but also challenged their own apprehension of sitting the Leaving Cert. Instead of the term "weak", my new understanding of intelligence and social class would rather describe students like him as struggling. In my opinion, the student's difficulties were a result of his place within a contradictory meritocratic system that puts blame on the student rather than itself as a broken system. The fact that he began to overcome these difficulties through extra one-on-one tutoring that he normally wouldn't be able to afford shows again that social class can have an unfair impact on achievement. Highlighting examples like private education and extra-curricular activities, Lynch and Cream (2018) explain that Irish children from poorer households cannot engage with the education system on equal terms with others as they do not have equal resources in a time of rising economic inequality. They describe the idea of meritocracy as "an unrealizable myth in an economically unequal society".

Burns (2014, p.9) has advised that implicit cultural assumptions held by teachers in DEIS schools, like the validity of a meritocratic system or the validity of fixed ability, results in "normalisation of low student attainment" and a "self-imposed achievement ceiling in relation to what they as educators can achieve". By challenging those assumptions in this issue, I feel this experience has emphasised to me the importance of my own values and efforts as a teacher toward maximising every student's achievement as much as I can.

Conclusion

Research into the effectiveness of teacher education programmes has commented on “the lack of impact teacher education has on students’ beliefs about teaching and learning and their later practices in schools” (Klein, 1998, p.1). Many Irish teachers feel that “research is of little practical value to them” (Gleeson, 2012, p.1), and that most of their skills “were acquired in classroom and school settings rather than during initial teacher education” (Gleeson, 2012, p.12). After carrying out this assignment, my own opinion is that educational theory and reflection can be combined in a way that leads to powerful conclusions that can have long-term effects on my teaching. Making sense of issues involving identity, inclusive teaching, meritocracy and social class has motivated and excited me about the possibilities of teaching and I endeavour to keep up with research as I continue in my practice.

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ED303: Opening Statement

1. Learning and Teaching philosophy

Teaching practice is a foundational aspect of a pre-service teacher's career, providing essential practical experience where students can learn from experienced teachers and develop key skills "particularly in the areas of planning, teaching, classroom management, and the organisation of learning activities for pupils" (Department of Education and Science, 2006, p.1) These are all skills that I hope to develop during my teaching practice, as I transition from identifying as a university student to a pre-service teacher. Putten et al. (2014) explain that a teacher's professional identity is an important factor in determining how learning and teaching take place in a classroom. Morris & Glass (p.2) advise that teachers first begin learning to teach "by growing up in a culture – by serving as passive apprentices for 12 years or more when they themselves were students". They warn that when pre-service teachers first face the challenge of transitioning from university lectures to teaching their own class, "they often abandon new practices and revert to the teaching methods their teachers used" (Morris & Glass, p.2). Based on this, I feel it is important to clearly understand my own identity and beliefs as a teacher, by exploring how my years as a student have influenced me, and what research and theories of learning I feel are important.

My teaching philosophy and values have been heavily influenced by my time in school, with my experiences in primary and secondary school being very similar to each other. Both schools were Irish-speaking, with a predominantly middle-class student population. There was a meritocratic ethos and all teachers had a formal traditionalist approach to their teaching. The traditional teaching style is something that happened to suit me as a learner. I found that I easily understood my teachers' explanations and looked forward to spending time practicing procedural tasks and applying the latest methods they gave me. I easily understood the abstract explanations and had no difficulty interpreting my teachers' traditional style. This teaching style, however, didn't seem to fit with other students in my classes. If they couldn't understand my teachers' notes, little time was given to offer alternative explanations and many students were simply seen as not being 'maths people'. My time spent in university pursuing this BME has given me an educational context to better understand my time in school. Reflecting upon my school experience has helped me realise that much of my own learning has been dominated by a traditional 'talk and chalk' approach to mathematics, one that expected the students to replicate procedures followed by the teacher regardless of whether the student even understood what they were doing. As a Leaving Certificate student, I struggled to understand why there was an expectation that struggling students should be able to afford tutoring outside of school and spent much of my time trying to help friends in my class that were struggling. I can see that a big part of my motivation to become a teacher was in the hopes of making a difference, teaching in a way that students of all ability levels . Since my time in secondary school, the introduction of the Project Maths syllabus has highlighted the importance of a more modern approach to the teaching of mathematics. In contrast with the traditionalist style of teaching I experienced, a modern constructivist approach is centred around students' understanding and is an approach that I hope to research and utilise in my own classroom.

Constructivism is a broad term that encompasses the work of many educational psychologists, including Jean Piaget and Lev Vygotsky. While there is no single constructivist theory of learning, constructivists are said to share two main ideas: that learners play an active role in constructing their own knowledge, and that social interactions heavily influence knowledge construction (Bruning et al., 2004). In contrast with how traditionalists see learning as a passive transfer of knowledge, psychological constructivists propose that an individual's knowledge, identity and beliefs all influence how learning takes place (Woolfolk et al., 2013). Piaget (1965) advises that children advance through four stages of cognitive development, with their thinking at each stage incorporating those previous and forming a personal cognitive structure. He goes on to explain how the processes of assimilation and accommodation are used by the learner to process new information. Assimilation occurs when the learner interprets new information in the context of their current structure of cognitive development, modifying it to fit with existing knowledge and beliefs. Accommodation, on the other hand, occurs when existing cognitive structures are adapted or updated to account for the new knowledge (Piaget, 1985). A consequence of this, since learners progress through the cognitive stages at their own rate, is that learners' understanding can be limited by how they've progressed through the different developmental stages. Acknowledging that each learner constructs their own understanding at different times, Piaget's theory of Cognitive Development highlights the importance of differentiated instruction and being aware of learners' current understanding. Piaget's work has helped inform my approach to teaching. For example, I always try to revisit the prior knowledge needed for a maths lesson, as students who have had previous difficulty in understanding might benefit from a refresh. I try to ask the students what they've had difficulty with rather than assume that they learnt it previously. Learners that were previously unable to grasp certain concepts when they first encountered them might now have progressed through the stages of cognitive development and be better able to understand, contextualise and accommodate the knowledge.

Lev Vygostsky, a social constructivist, proclaimed that learning is shaped by "social interaction, cultural tools and activity" (Woolfolk et al., p.404, 2013). Through participating in activities with others, learners internalise and appropriate the knowledge and strategies of those around them. Vygostsky (1962) believes that learning takes place inside the 'Zone of Proximal Development'. In this zone, learners can master concepts they wouldn't fully grasp on their own by working together with others that are more advanced. Learning in this zone takes place through intersubjectivity and scaffolding. Intersubjectivity defines the process that occurs when two learners who begin a task with different levels of understanding grow towards a shared understanding through communication and adjusting to each other's perspectives. Scaffolding, on the other hand, involves the teacher adjusting the level of their help to best fit the current understanding of the learner and maximise their learning (Woolfolk et al., 2013).

Both Piaget and Vygotsky highlight the importance of viewing the teacher as a facilitator of a student's active construction of knowledge, rather than the traditionalist practice I experienced in school. In my opinion, in order to be an effective teacher an effort should be made to account for the diversity of any given classroom. Content should be adapted so that it is taught in the context of real life applications, and lessons should be designed in a way to promote student discussion and foster their natural curiosity for learning. After outlining the context of the school where I will be undertaking placement in Section 2, I will highlight how these theories of learning have informed the learning goals of my teaching practice and the methodologies I hope to incorporate.

2. School Context

My teaching practice will take place in a local post-primary school in Galway City that facilitates 300 students. These students are of diverse socio-economic and cultural backgrounds, and the school prides itself in promoting both academic and vocational achievement. I will have 2 roles in the school, one as a mainstream teacher teaching a single class, and another as a resource teacher during after-school study.

My mainstream class is with the school's Transition Year students, where the class is a mixture of higher level and ordinary level students. The class has 22 students, attendance fluctuates wildly, however. Among these students are 4 English as an Additional Language students. In communication with my co-operating teacher, the maths topic of my classes will be decided weekly based on what the co-operating teacher is currently teaching. In terms of resources, the students don't have a textbook assigned to them for Transition Year and lessons will have to be designed around the use of preparing materials in advance as well as utilising the whiteboard and projector available in the classroom.

My role as a resource teacher will involve spending two hours per week supervising the school's Evening Study. This is a service provided free of charge and supports students from 1st Year to 6th Year by providing them a comfortable place complete their homework or study. I hope to use this time to support students with their homework or study, and I see it as an opportunity to help improve students' learning in other classes by helping them with any work they're currently struggling with.

3. Learning Goals for Placement

I view this teaching practice as a solid foundation for my future career as a teacher. In this sense, it’s important to me that my teaching is informed by research. I have previously outlined educational theory that I feel has improved my understanding of student learning, and I hope to design my lessons in a way that facilitates this. My goals are to:

- Utilise problem-based learning, to promote discussion in class and move away from a teacher-centred approach.
- Identify differentiated learning goals to make sure each lesson is designed in a way to support all students.
- Make use of mixed-ability grouping, to promote peer learning through intersubjectivity.
- Improve my questioning skills and integrate scaffolding into my practice.

It’s important to plan different actions throughout my placement to ensure I’m working towards these goals:

- Lesson Planning. Without a lot of time spent planning lessons, I won’t be able to create the type of learning environment that I’ve set out in my goals above. I will make use of high-quality resources, as well as skills developed in other modules such as ‘Mathematics Methodologies & Skills of Teaching', to try to design lessons as best I can.
- Reflective Practice. As a pre-service teacher I believe that I have a lot to learn, and in order for me to benefit from research-informed practice I need to be able to see how my efforts translate to the classroom. By undertaking weekly reflections, I hope to learn from any successes or problems that might arise. This will be a consistent way for me to understand why different ideas did or didn’t work and help me improve my practice for the future.
- Classroom Observations. My co-operating teacher has advised that I can observe her other classes when our timetables allow, which is something that I will be sure to do. Learning from a more experienced teacher and seeing how they manage classrooms and design lessons will be an invaluable resource during placement.

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Critical Incident Report

The ability to critically reflect has been described as a defining characteristic of any professional practice (Schön, 1983; Walsh & Dolan, 2009). Walsh & Dolan (2009, p.139) suggests that when we observe expert teachers and their actions, we miss the "fundamental dimension" of their internal thought process. From the outside, expert teachers seem to instinctively know how to adjust lessons in the face of surprises, for example. While their actions could be studied and reproduced, Nesbit et al. (2004, p.74) explains that the most valuable skill of a great teacher is their "deep understanding of themselves and their students, and of the organisational contexts in which they work". This deep understanding is something that can be developed through reflection, both in practice and on practice (Schön, 1983). As a pre-service teacher aspiring to be one of those great teachers someday, I hope to develop my critical reflection skills to the best of my ability. Focusing on reflection on action, Walsh & Dolan (2009) advise multiple strategies to promote critical reflection, including keeping a reflective journal and carrying out a critical incident analysis. I aim to carry out a critical incident analysis in this assignment, and will be using the reflective journal that I kept throughout this teaching placement to do so.

In their description of critical reflection, Schön (1983, p.68) has said that its important for a practitioner to allow themselves to feel "surprise, puzzlement, or confusion in a situation which he finds uncertain or unique". Through reflecting on the prior experiences that have informed their "implicit behaviour", the practitioner then "carries out an experiment which serves to generate both a new understanding of the phenomenon and a change in the situation" (Schön, 1983, p.63). From this, I can see that it is important to develop a cycle of reflection and action, each informing the other. Using my reflective journal, I have identified a situation that left me feeling uncomfortable and unprepared as a teacher, and I hope to understand this better in order to inform my future practice.

This incident occurred during my Resource Teaching at my DEIS placement school's After-School Study Program, at the beginning of the year. Students are typically allowed to work together to do their homework and on this day a pair of Leaving Cert boys were working together on their Maths homework. Their conversation would fluctuate between the work at hand, and jokes made to each other. Despite them sometimes talking off-topic, they were still focusing on their work so I let them talk as long as they weren't too loud. After a few minutes of answering questions about Financial Maths, they started commenting and joking to each other about their own ethnicity and social class. Using words like “gypsy” and “African”, they didn’t think either of them would be able to use Financial Maths to “fix being poor”, “especially going to this school”. After their quick back-and-forth they refocused on their work, and I chose to ignore their comments as I was worried I would re-ignite their conversation by drawing attention to their comments. I felt out of place, in both my own ethnicity and my educational background. When similar comments happened again, I advised the students that I’d be happy to help them with their Maths but they aren’t permitted to talk to each other anymore as they’re being too loud.

This incident stood out to me as important to reflect on as it prompted me to write multiple questions to myself in my reflective journal: the students discussing topics of racial ethnicity and social class made me feel uncomfortable, why is this? What kind of language is okay in school, am I supposed to intervene if a student draws attention to their own ethnicity? How does that kind of behaviour fit into promoting an inclusive environment? Did I react in the right way?

Upon reflection, I can see that the students discussing their own ethnicity, social class and school in a negative light left me feeling uncomfortable at the thought of speaking up. This uncomfortableness stemmed from a sudden realisation of a stark contrast between my own educational background and that of these students. Although I am a working class student myself, I attended both an Irish-speaking primary and secondary. The schools I attended served a predominantly white, Irish, catholic population of students. In fact, during my time in school I never interacted with a student who wasn’t white and Irish. I can see now that overhearing students discuss their own ethnicity made me feel inadequately prepared to express my opinion on what they were saying, as I’d never even encountered those kinds of topics in during my own school experience. I feel this incident is significant in the context of my place in the education system as a whole. Over the past two decades, the student population of Irish schools has been growing increasingly culturally diverse, especially in DEIS schools (Devine, 2005; Burns, 2015). In contrast, the teaching population in Ireland has been described as homogenous in its high-level of white, Irish teachers and “is less culturally and ethnically diverse than in other OECD countries” (Hyland, 2012, p.10)

Initially, I felt like I dealt with the situation well considering how conflicted I felt at the time. I didn’t feel like it was my place as a white student-teacher, who attended a school where a high-percentage of students progress to third-level education, to tell students who have different cultural and educational experiences how to feel or act. This led me to feel justified in simply telling the students to stop talking, rather than bring up the topics they were discussing, which I felt they were more experienced in. Upon reflecting on my role as a teacher, as well as trying to understand the point of view of the students, I no longer feel that my reaction was right.

By noting it in my reflective journal, this incident was something that was in the back of my mind as I was engaging with other parts of the course, and certain topics I learnt in my 'Psychology, Sociology and Inclusive Teaching' helped me come to a deeper understanding of it. Regardless of a teacher's background, it's important to promote a positive and inclusive environment for all students. Intercultural education is of growing importance in our education system, and involves not only celebrating and recognising the diversity of students but also challenging negative bias (NCCA, 2006)

The NCCA (2006, p.41) suggest steps for teachers to take in situations where negative situations involving diversity arise. One of the first things they suggest is to take immediate action and not to wait "to see if the behaviour will change on its own". This suggestion has helped me feel more comfortable in my initial response, in the sense that it is a common response for teachers in those situations. The NCCA (2006, p.41) emphasise the importance of challenging negative bias, and emphasise the need to make clear that "certain behaviour or responses are inappropriate". With this in mind, I can see now that my personal values and intentions were not aligned with my actions during this critical incident. As I was listening to the students putting themselves down, I had strong opinions about the importance of a growth mindset, positive self-concepts and self-efficacy and how they relate to academic achievement (Bandura, 1997; Dweck, 1999). These constructive opinions were overshadowed by how I was uncomfortable with intervening, and ultimately lead to inaction. I can see now that by only asking the students to stop talking, and not challenging their comments or engaging with them, it could even be interpreted by the students that I agree with them or that their behaviour was expected. I don't want to be in a situation again where students negative views of themselves go unchallenged, all students deserve to hear positive encouragement and engagement from their teacher, and I feel like that is how I would approach this situation again in the future.

Reflecting on this incident gave me motivation to research further, and after studying the guidelines put forward by the NCCA, I feel much more comfortable with the topic at hand.

Regardless of my own personal background, I understand the value of promoting an inclusive and intercultural environment, challenging negative views of students in favour of promoting positive ones. I expect my future as a teacher will involve similar situations again, but I plan for my response to be different. I now understand the importance of putting theory into action, and feel more excited to do so. My views on this situation have grown from seeing it as a 'problem' that I wanted to ignore, to seeing it as an opportunity for both me and the students – an opportunity for me to put valuable theory into practice. Burns (2015) has highlighted a lack of "engagement with and a valuing of diversity" as a challenge for DEIS schools, and I feel like this incident was a missed opportunity where I could have expressed the value of diversity directly to the students. Thanks to this reflection I've been able to recognise this, and aspire not to simply ignore similar situations in the future.

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Closing Statement

a) Review of learning and teaching philosophies and goals set at the beginning of and throughout the year

The most prominent goals that I set at the beginning of the year were related to moving away from the traditional style of teaching that I experienced in school, and undertaking a more inclusive and constructive approach to planning lessons. Researching the constructive approach, I identified the importance of utilising problem-based learning, differentiated learning goals, mixed-ability grouping and peer learning. I also hoped to develop my questioning skills and my ability to integrate scaffolding into my practice.

In order to recognise my learning throughout the year, I expected to see learning and improvement reflected in each lesson plan that I prepared. I also placed an emphasis on reflective practice and decided to undertake a reflective journal.

b) Review of learning and development (addressing overall challenges, successes, personal and professional growth and learning)

Writing this closing statement at the end of my placement, I am glad to be able to identify personal and professional growth after overcoming challenges and improving my teaching ability throughout the year.

One of the most challenging experiences of the year, also gave way to one of the aspects of the year that I value the most. At the beginning of the year, I was overcome with anxiety on the morning of my first lesson. This stemmed from feeling out of practice in terms of teaching, and feeling daunted by preconceived opinions of teaching in a school that was 'difficult'. This battle with performance anxiety is something that I've improved greatly with throughout the year, and a big factor in this improvement is the relationship I developed with my co-operating teacher. Sensing my anxiety on my first day of teaching, my co-operating teacher went out of her way to encourage me not to give up, sharing her experiences of teaching and explaining that other teachers experience similar feelings.

I feel like my anxiety effected my teaching performance negatively at the beginning, I was pre-occupied with controlling the behaviour of the class and not adequately reflecting on my own performance. Building on my relationship with my co-operating teacher, I observed one of her classes once a week. This was an invaluable experience, it allowed me to see that even experienced teacher like her faced similar challenges. A key difference, however, was her ability to persevere through difficult interactions with students. These observations, and my discussions with the teacher after each class, inspired me to improve my practice. I no longer ascribed difficulties in class to personal deficits, and saw instead different ways that I could improve my practice. This growing confidence translated into a more positive atmosphere in class, and the students responded to my positivity with their own. By the end of the year, I found myself growing more comfortable moving away from traditionalist teaching strategies, allowing students more freedom and designing lessons in more interesting and effective ways (e.g. Lesson 15 – student-led statistical investigation).

My strong feelings of anxiety returned at the end of the year during my teaching inspection. I expected very bad results, due to a lack of confidence. Thanks to my professional development throughout the year, however, I was told that my lesson was well-designed, that I showed effective questioning skills, and that I had a "great aura about" me while teaching. The main criticism was that I should work on my confidence. This is something that I can see has improved throughout the year and I look forward to improve on it in the years to come, by continuing to focus on carry out a reflective practice.

c) Reflection on your experiences as it relates to other modules and to the BA in Mathematics and Education as a whole.

Now that the year is over, I especially value this practical teaching experience for the challenging experiences I had in class. These experiences led me to try to find solutions in other modules, and I found very applicable research in my 'Psychology, Sociology and Inclusive Education' module. Important concepts like those of meritocracy, inclusive education, self-efficacy, and social class identity not only helped me overcome the challenges that I met in the classroom but also in my own academic performance. I have a reinvigorated appreciation for the theory that we learn, and am excited to continue to put it into practice in order to improve my own learning as well as that of my future students

Assignment 1: Educational Autobiography **Cian Mac Liatháin - 11476078**

1.

As a 21 year old, Caucasian, middle class, settled Irish male I am not part of any social group or minority that is prejudiced against or at an obvious disadvantage to others. For this reason I have always considered my educational experience to be ‘normal’ but due to the self-reflection necessary to carry out an Educational Autobiography I am now able, to identify the influence that both my social class and personal family life has had on my educational experience.

The middle-class, Caucasian, settled-Irish characteristics of my family meant that my older brother (3 years my senior) and I were able to attend our local Jesuit bunscoil. Despite its selective admission policies, the school had a strong reputation and its plentiful funding meant that there were ample resources for every student to receive a high quality education. This is in contrast with schools only down the road, where the children of other ethnicities/social classes would attend.

While I was in 6 years old, my parents divorced and they both were quick to re-marry. My brother and I would split our time equally between my grandparents’ and each of my parents’ homes.

I can see now the influence this had on me as a child, as my brother and my classroom were the most consistent aspects of my day-to-day life. This resulted in me mimicking my brother’s views and opinions as my own, as well as becoming introverted and putting a great deal of effort into perfecting my schoolwork from a young age.

Despite my immediate family having no history of third level education, my brother placed great value on it at a young age. At the age of 12, my brother insisted we be enrolled in our local meánscoil lán-Ghaeilge; he recognised that education through the Irish language often goes hand-in-hand with education of a high quality, which then leads to 3rd level. It’s fair to say that this kind of late enrolment wouldn’t have been possible but for my family’s social standing and pre-existing relationships with some of the school staff – sociological factors that shouldn’t, but unfortunately do, have an effect on the quality of education available to most children.

Through hindsight, I can acknowledge the extent of which I mimicked my brother’s views and opinions throughout my school life, ignoring my own educational ambitions. He believed that there was no place for any kind of assistance or community in the education system; everyone is equally capable and individually responsible for their own education, the only thing that stands between a student and the highest mark is putting in effort.

Thanks to the emphasis I placed on doing my own schoolwork to a high standard in primary school, I had a solid foundation in all subjects and found all aspects of school straightforward and easy to do well in. After going through secondary school finding all the subjects easy to comprehend and memorise, I slowly began to take note of how other students were doing around me. I was disappointed to see teachers doing the bare minimum and leaving most of my classmates struggling.

I finally felt obliged in 6th year to form my own view of the education system, which turned out to be nearly polar opposite to my brother’s: I don’t believe that everyone is individually equally capable of greatly organising their entire education, but I believe that a great teacher can ensure that everyone is at the level they need to be, and if they can’t then it’s a shortcoming of the teacher not the student.

Personally, I was confident that I would get the points needed for my course, so felt that any more study for my own subjects would be pointless since most of my class weren't lucky enough to be in a situation where they would have surplus points. So I spent my 6th year helping my classmates in any way that I could. I learnt the syllabus for subjects I'd never done in order to give grinds, helped with projects and portfolios that needed to be finished etc. In the end I got 500 points, and made a significant difference in 20 people's Leaving Cert, in comparison to my brother's 800 points.

Through the undertaking of this Critical Educational Autobiography, I can see what individuals, event and sociological characteristics have influenced my educational experiences and aspirations. I can now clearly identify the reasons for which I'm passionate about becoming a Maths and Irish teacher, the two subjects that students find most difficult.

2.

In my opinion there are three variables that determine an individual's success or failure in school: home life, the student's own personality and the quality of their teachers. Certain home situations can leave students disadvantaged when it comes to attaining success in school. In general the individual's personality must be able to understand the reasons to succeed in school and the consequences of failing, and be willing to put in whatever effort is needed. If there is a situation at home which leaves them disadvantaged then this mental strength needs to be even stronger, so as to be able to prioritise and separate the school and home life from each other.

In my opinion, a teacher should be able to inspire students to this required way of thinking and help the students with external pressures/problems all while effectively teaching the required syllabus. Unfortunately, for reasons such as under-resourced schools, rising class numbers, underpaid teachers etc., all teachers aren't held to the standard of 'great' teachers and I think that is a big factor in why individuals fail or succeed in school.

Cian Mac Liatháin – 11476078 **Assignment 1: Psychology and Development**

This assignment will focus on cognitive development, and the two popular approaches used to understand it: the biological aspects of the brain, and Piaget's theory of development.

Due to advances in modern science, scientists are now able to observe brain activity in response to specific actions e.g. speaking a second language or solving a math problem. These advanced scientific capabilities have allowed for much more research to be carried out on the brain and its direct role in cognitive development in the past. (Bransford, Brown & Cocking, 2000)

In contrast to this direct observation of the brain, Swiss psychologist Jean Piaget carried out ground-breaking research through the observation of children themselves, and used these observations to devise a theory of cognitive development. (Woolfolk, Hughes and Walkup, 2008, p.38). Described as "the most influential developmental psychologist in the history of psychology" (Slavin, 2009, p.31**).** Piaget had a doctorate in biology, and applied biological principles to the study of cognitive development. This may explain why, despite using fundamentally different methods of observation, these two approaches share core beliefs that are heavily debated between developmental psychologists (Rathus, 2010):

1. Nature – Nurture

Researchers have been attempting for some time now to determine the extent to which human behaviour is the result of heredity (nature) or environment (nurture). What behaviour is as a result of our genes and the unfolding of our biological programming, and what behaviour is as a result of experience and learning? Today, most psychologists agree that aspects of nature and nurture combine to influence our development.

In terms of the brain, neuroscience has taught us the intricacies of how information is communicated. Neurons are cells that "send out long arm-and branch-like fibres to connect with other neuron cells and share information by releasing chemicals that jump across the tiny spaces between them, called synapses" (Woolfolk et al., 2008, p.32). The number of neurons formed at birth is the amount we have for life – around 100 billion**.** This information highlights that genetics (nature) are responsible for the basic wiring of the brain (the forming of the neuron and the general connections between regions of the brain). In addition, Santrock (2009**)** explains that experience and learning (nurture) are responsible for calibrating these connections, "helping each child adapt to the particular environment (geographical, cultural, family, school, peer-group) to which he belongs".

For example, our brains are formed with the basic circuitry to learn language – we can recognize speech, identify subtle differences and understand the grammatical rules for ordering sentences. However, the language which is learnt, the extent of vocabulary, the accent and dialect are all determined by the environment in which the child is raised.

Similarly, Piaget's theory attributes cognitive development to a combination of nature and nurture: "One of the most important influences on the way we make sense of the world is maturation, the unfolding of the biological changes that are genetically programmed." (Woolfolk et al., 2008, p.38) This maturation grants the ability to perceive, learn and act.

With regards to nurture, Piaget's theory also highlights two common tendencies between species: organisation, assimilation and accommodation. Organisation describes how children integrate particular observations into a body of coherent knowledge. For example, young infants have difficulty looking at and grasping an object simultaneously. As they develop, they organise these two separate actions into a 'scheme' which allows them to look at, reach and grab an object can (Slavin, 2009). Assimilation and accommodation are adjustments in response to the environment of the child. Assimilation involves adding new information to a scheme and using that to understand something new. Accommodation takes place when schemes are altered or new ones are made in response to new information.

2. Active – Passive

Another topic that developmental psychologists debate over is that of whether children are active or passive learners. That is, whether children actively shape and direct the course of their development or whether they are simply passive recipients of experience. Many educators agree that individual differences can be seen between children and that their activity or passivity can change from subject to subject (Rathus, 2010).

In general however, it can be shown that both the study of the brain and Piaget's theories agree that children are active learners. According to Slavin (2009, p.171) "one cannot alter a brain that is not yet ready for incoming experience to affect it". A brain must biologically progress to a stage that it allows its environment to alter it, "change through learning cannot exceed the developmental status of the neural structure."

Piaget details that physical maturation brings along with it an increasing ability to act upon the environment and learn from it. For example, a child whose coordination has reasonably developed can discover the principles of balance by experimenting with a seesaw. Thus, as we explore our environment and organise that information we alter out thinking processes and learn at the same time. (Woolfolk et al., 2008)

3. Continuity – Discontinuity

Whether our learning is a continuous process or happens in separate stages is a concept with which our study of the brain and Piaget's theory do not agree.

Although the number of neurons we have does not grow after birth, the amount of synapses (the links between neurons) does. Throughout our lives, synapses are oversupplied and then 'pruned' as a response to our environment, to meet our needs (Bransford, Brown and Cocking, 2000). Although there are periods in life that our brain is more sensitive to certain experiences and can produce more synapses accordingly if triggered (experience-expectant), stimulus during these stages is not mandatory and can be made up for, albeit with more difficulty (experience dependent) (Woolfolk et al., 2008).

In contrast to this quantitative view, Piaget's theory divided our cognitive development into four distinct stages: sensorimotor, preoperational, concrete operational, and formal operational. Piaget believed that "all children pass through these stages in this order and that no child can skip a stage"

(Slavin, 2009, p.33) Each stage builds upon the abilities learned through previous stages, for example the ability to solve abstract problems in the formal operational stage (11 years – adult) stems directly from the use of memory and thought that first begun in the sensorimotor stage (0 – 2 years).

Implications for Education

Using the insights highlighted above, it is possible to see what implications each of the approaches to cognitive developments may have.

The continuous and active development of the brain in response to our environment has many implications for education. An active and diverse environment and flexible methods of teaching in younger years would provide a large group of children with a wide range of cognitive development. Over time, students are likely to have different preferences with regards to modes of processing information (visual learners, aural learners etc.) and in later years teachers should be especially conscious of the required diversity in the methods of teaching.

The discontinuity of Piaget's theory has obvious implications for education. Each of his stages outline what development the brain will be undergoing and education should respond accordingly, For example, the preoperational stage (2-7 years) is characterised by the development of language and language development should be a focus of children's education at this age.

It is evident that education should be closely linked to the research and study of the brain and children. The two approaches above, however, only represent a small portion of our knowledge and theories. I hope over time that our resources and research can combine in order to have a more efficient education system.

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